

Data 260

Homework 3

Name: Kshitija Shinde

SJSU ID: 019146491

Value	Definition
SID4	6491
PORT_BASE	$8000 + (6491 \bmod 900) = 8000 + 191 = 8191$
PREFIX	"s" + 6491 = s6491
SEED	6491
VERIFY_SEED	$260000 + 6491 = 266491$
DOMAIN_ID	$6491 \bmod 8 = 3$
Assigned Domain	Grocery supply and recall notices

Hardware: MacBook Air, 13-inch, M4, 2025, Apple M4 chip, 16GB RAM, macOS Tahoe 26.5.2

Local model used: qwen3:8b via Ollama

Tagged commit hash: 7077b84120e75ab4dbd45b99b70995bf5fc89405

Repository: <https://github.com/Kshitija-Shinde9/data260-6491>

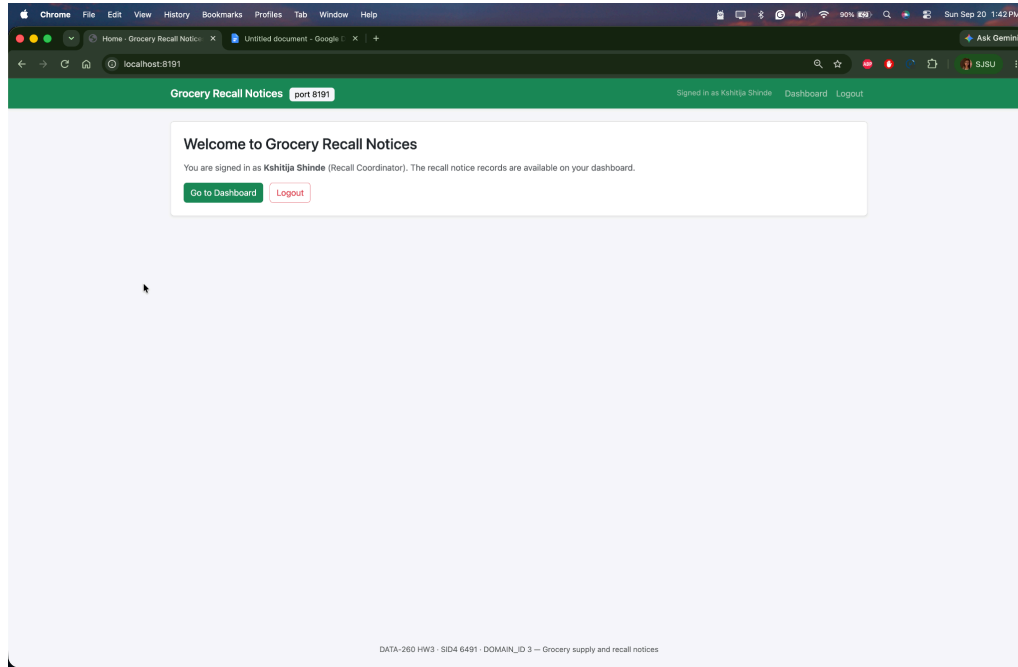
Objective: Create a FastAPI application that implements a simple user authentication system for your specific domain. The application should include:

- User login and logout functionality
- Session management to keep users logged in
- Protected routes that only logged-in users can access
- Styling using Bootstrap to make the application visually appealing

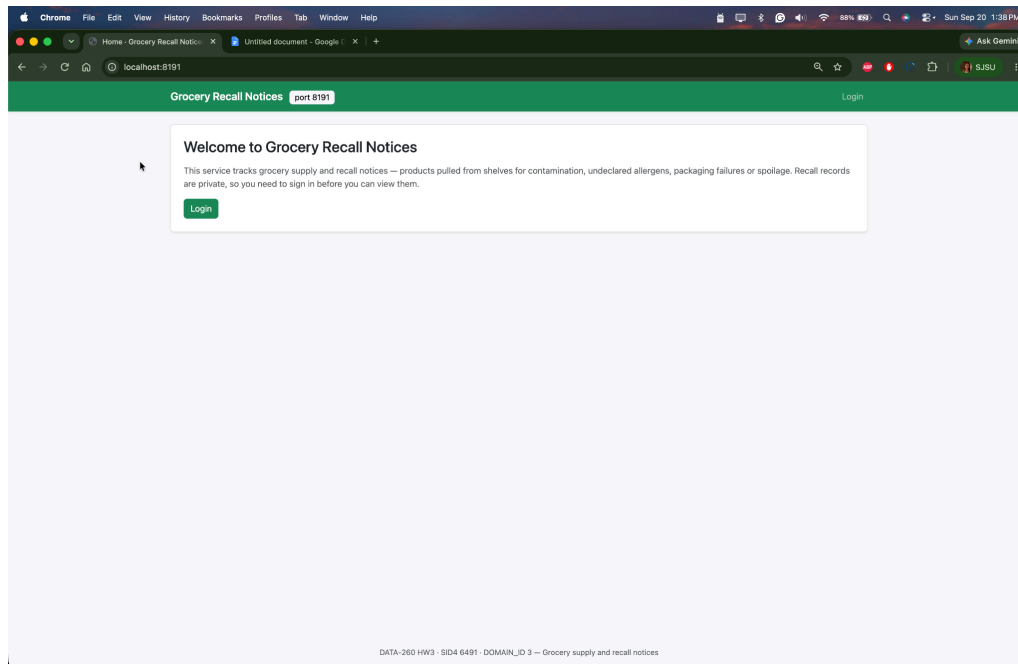
Routes - Handled Separately in a router:

Home Page (/): Display a welcome message for your domain application. Show a link to the login page if the user is not logged in. Show a link to the dashboard and log out if the user is logged in.

Screenshots - With Login



Without Login

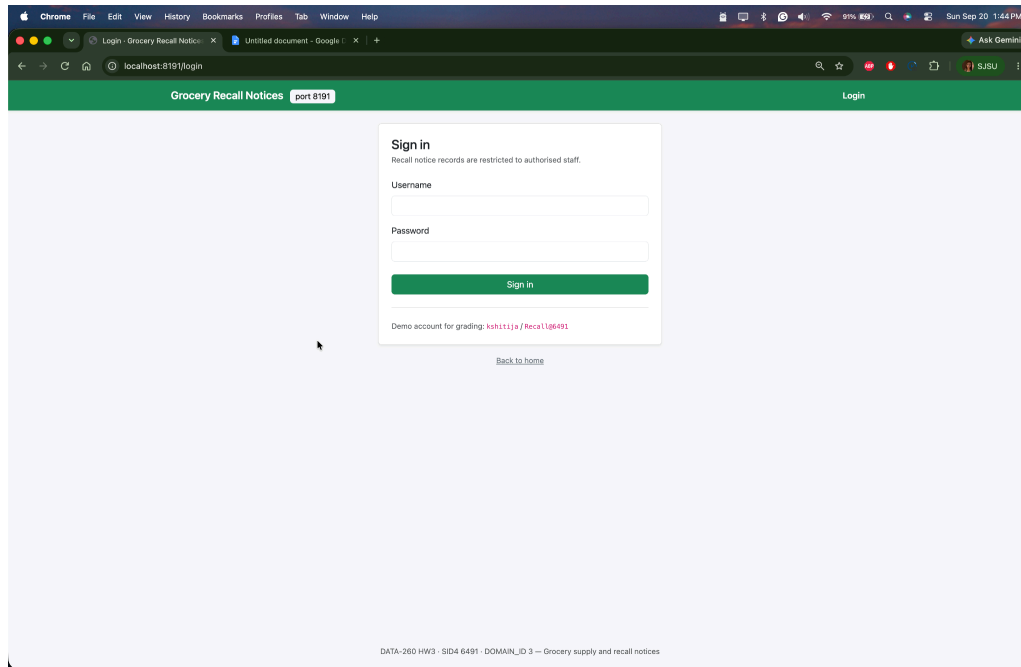


Code -

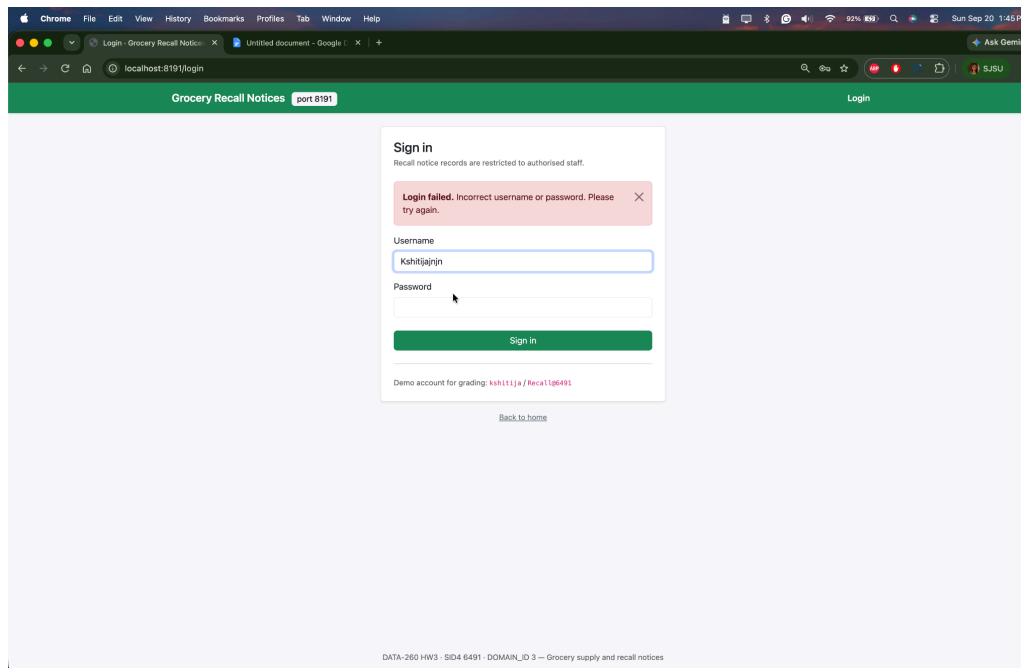
```
<> index.html U X
code > web_application > templates > <> index.html > ...
1  {% extends "base.html" %}
2
3  {% block title %}Home &mdot; Grocery Recall Notices{% endblock %}
4
5  {% block content %}
6
7  <div class="card shadow-sm">
8    <div class="card-body p-4">
9
10     <h1 class="card-title h3 mb-3">Welcome to Grocery Recall Notices</h1>
11
12     {% if user %}
13     <p class="card-text text-body-secondary">
14       You are signed in as <strong>{{ user.name }}</strong> ({{ user.role }}).
15       The recall notice records are available on your dashboard.
16     </p>
17     <a class="btn btn-success" href="/dashboard">Go to Dashboard</a>
18     <a class="btn btn-outline-danger ms-2" href="/logout">Logout</a>
19
20     {% else %}
21     <p class="card-text text-body-secondary">
22       This service tracks grocery supply and recall notices &mdash; products
23       pulled from shelves for contamination, undeclared allergens, packaging
24       failures or spoilage. Recall records are private, so you need to sign
25       in before you can view them.
26     </p>
27     <a class="btn btn-success" href="/login">Login</a>
28
29     {% endif %}
30
31   </div>
32 </div>
33
34 {% endblock %}
35
```


Login Page (/login): Display a login form with fields for username and password. Validate the credentials and log the user in if they are correct. Redirect to the dashboard on successful login. Display a Bootstrap alert message on the login page when a user enters invalid credentials.

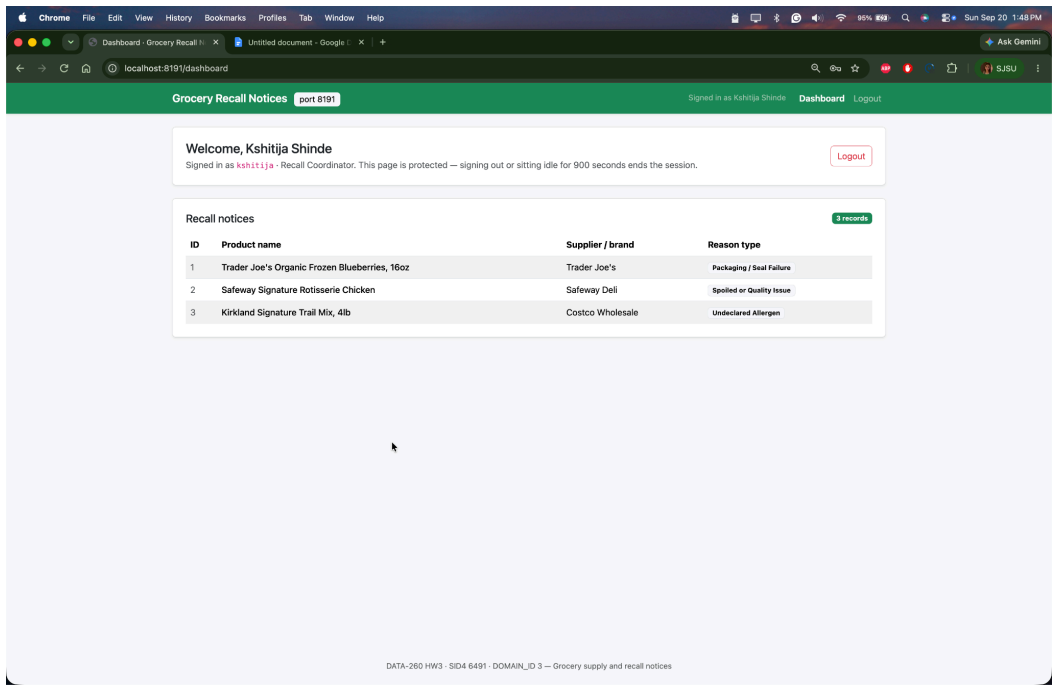
Screenshots - Login screen



Login screen with incorrect credentials



Login screen with correct credentials redirected to dashboard page

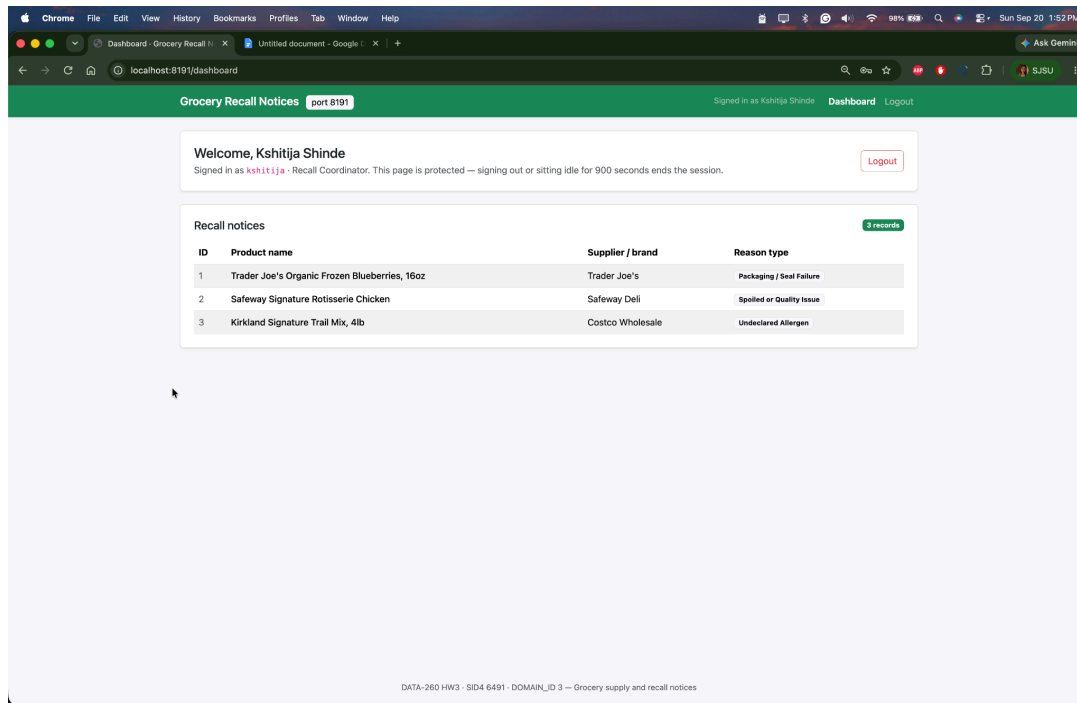


Code -

```
login.html U X
code > web_application > templates > login.html > div.row.justify-content-center > div.col-md-6.col-lg-5 > div.card.shadow-sm > div.card-body.p-4 > p.text-body-secondary.small.mb-4
1  {% extends "base.html" %}
2
3  {% block title %}Login &ndash; Grocery Recall Notices{% endblock %}
4
5  {% block content %}
6
7  <div class="row justify-content-center">
8    <div class="col-md-6 col-lg-5">
9
10     <div class="card shadow-sm">
11       <div class="card-body p-4">
12
13         <h1 class="card-title h4 mb-1">Sign in</h1>
14         <p class="text-body-secondary small mb-4">
15           | Recall notice records are restricted to authorised staff.
16         </p>
17
18         {% if error %}
19         <div class="alert alert-danger alert-dismissible fade show d-flex align-items-start"
20           role="alert">
21           <div class="flex-grow-1">
22             <strong>Login failed.</strong> {{ error }}
23           </div>
24           <button type="button" class="btn-close" data-bs-dismiss="alert"
25             aria-label="Close"></button>
26         </div>
27         {% endif %}
28
29         {% if request.query_params.get("expired") and not error %}
30         <div class="alert alert-warning alert-dismissible fade show" role="alert">
31           <strong>Session ended.</strong> Your session expired or was logged
32           out, so it can no longer be used. Please sign in again.
33           <button type="button" class="btn-close" data-bs-dismiss="alert"
34             aria-label="Close"></button>
35         </div>
36         {% endif %}
37
38         <form method="post" action="/login" novalidate>
39
40           <div class="mb-3">
41             <label class="form-label" for="username">Username</label>
42             <input class="form-control" type="text" id="username" name="username"
43               value="{{ attempted_username or '' }}"
44               autocomplete="username" autofocus required>
45           </div>
46
47           <div class="mb-4">
48             <label class="form-label" for="password">Password</label>
49             <input class="form-control" type="password" id="password" name="password"
50               autocomplete="current-password" required>
51           </div>
52
53           <button class="btn btn-success w-100" type="submit">Sign in</button>
54         </form>
55
56         <hr class="my-1">
57         <p class="text-body-secondary small mb-0">
58           Demo account for grading:
59           <code>kshitija</code> / <code>Recall@6491</code>
60         </p>
61
62       </div>
63     </div>
64
65     <p class="text-center mt-3 mb-0">
66       <a href="/" class="link-secondary small">Back to home</a>
67     </p>
68
69   </div>
70 </div>
71
72 {% endblock %}
```

Dashboard Page (/dashboard): Display a welcome message with the user's name. Show a logout link. Protect this route so only logged in users can access it.

Screenshots - Dashboard page



Code -

```
dashboard.html X
code > web_application > templates > dashboard.html > div.card.shadow-sm
1  {% extends "base.html" %}
2
3  {% block title %}Dashboard &mdash; Grocery Recall Notices{% endblock %}
4
5  {% block content %}
6
7  <div class="card shadow-sm mb-4">
8    <div class="card-body p-4 d-flex flex-wrap justify-content-between align-items-center gap-3">
9      <div>
10        <h1 class="card-title h1 mb-1">Welcome, {{ user.name }}</h1>
11        <p class="card-text text-body-secondary mb-0">
12          Signed in as <code>{{ user.username }}</code> &mdash; {{ user.role }}.
13          This page is protected &mdash; signing out or sitting idle for
14          {{ idle_timeout }} seconds ends the session.
15        </p>
16      </div>
17      <a class="btn btn-outline-danger" href="/logout">Logout</a>
18    </div>
19  </div>
20
21  <div class="card shadow-sm">
22    <div class="card-body p-4">
23
24      <div class="d-flex flex-wrap justify-content-between align-items-center gap-2 mb-3">
25        <h2 class="h2 mb-0">Recall notices</h2>
26        <span class="badge text-bg-success" id="countBadge">&hellip;</span>
27      </div>
28
29      <div id="loadingState" class="text-body-secondary py-3">
30        <span class="spinner-border spinner-border-sm me-2" role="status"
31          aria-hidden="true"></span>
32        Loading recall notices&hellip;
33      </div>
34
35      <div id="errorState" class="alert alert-danger d-none" role="alert">
36        <strong>Could not load the notices.</strong>
37        <span id="errorText"></span>
38      </div>
39
40      <div id="emptyState" class="alert alert-secondary d-none" role="alert">
41        No recall notices have been reported yet.
42      </div>
43
44      <div class="table-responsive d-none" id="tableWrap">
45        <table class="table table-striped table-hover align-middle mb-0">
46          <thead>
47            <tr>
48              <th scope="col">ID</th>
49              <th scope="col">Product name</th>
50              <th scope="col">Supplier / brand</th>
51              <th scope="col">Reason type</th>
52            </tr>
53          </thead>
54          <tbody id="recallRows"></tbody>
55        </table>
56      </div>
57    </div>
58  </div>
59 </div>
```

```
dashboard.html X
code > web_application > templates > dashboard.html > div.card.shadow-sm
51      <th scope="col">Reason type</th>
52    </tr>
53  </thead>
54  <tbody id="recallRows"></tbody>
55 </table>
56 </div>
57
58 </div>
59 </div>
60
61 {% endblock %}
62
63 {% block scripts %}
64 <script>
65   async function loadRecalls() {
66     const loading = document.getElementById('loadingState');
67     const errorBox = document.getElementById('errorState');
68     const emptyBox = document.getElementById('emptyState');
69     const tableWrap = document.getElementById('tableWrap');
70     const badge = document.getElementById('countBadge');
71
72     try {
73       const response = await fetch('/api/recalls');
74       if (!response.ok) throw new Error('the server responded with ' + response.status);
75
76       const recalls = await response.json();
77       loading.classList.add('d-none');
78       badge.textContent = recalls.length + ' record' + (recalls.length === 1 ? '' : 's');
79
80       if (recalls.length === 0) {
81         emptyBox.classList.remove('d-none');
82         return;
83       }
84
85       document.getElementById('recallRows').innerHTML = recalls.map(r => `
86         <tr>
87           <td class="text-body-secondary">${r.id}</td>
88           <td class="text-medium">${r.product_name}</td>
89           <td>${r.supplier}</td>
90           <td><span class="badge text-bg-light border">${r.reason_type || ''}</span></td>
91         </tr>`
92       ).join('');
93       tableWrap.classList.remove('d-none');
94     } catch (err) {
95       loading.classList.add('d-none');
96       document.getElementById('errorText').textContent = err.message;
97       errorBox.classList.remove('d-none');
98       badge.textContent = '';
99     }
100   }
101
102   loadRecalls();
103 </script>
104 {% endblock %}
105
106
```

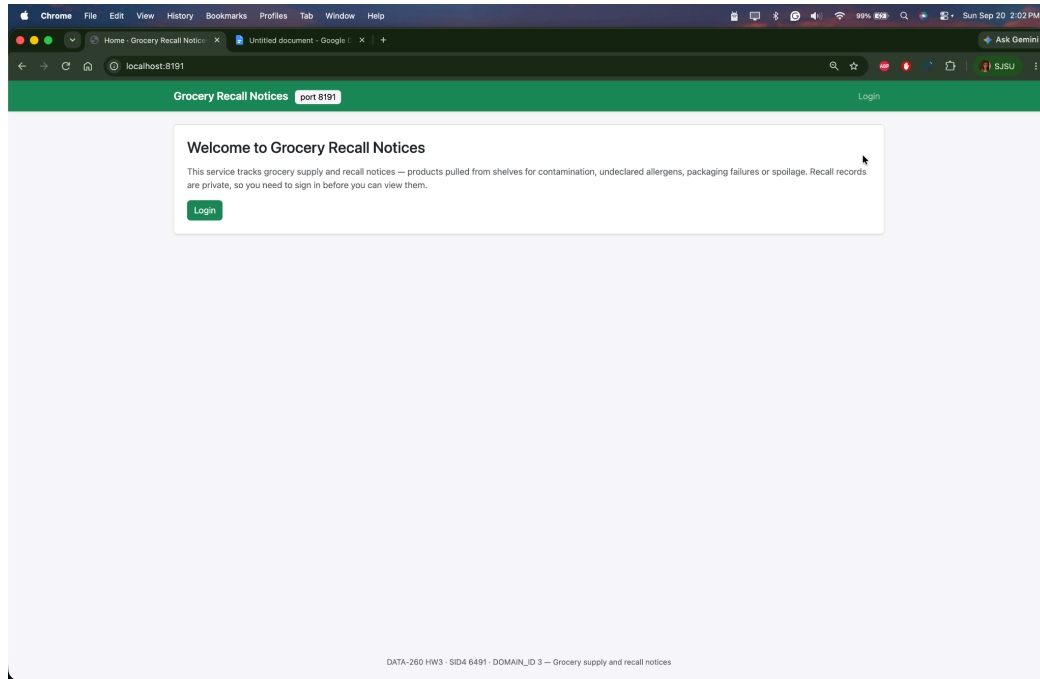
Code which shows that route which shows the dashboard is protected.

```
auth.py 5, U X
code > web_application > routers > auth.py > login_page
148
149 @router.get("/dashboard")
150 def dashboard(request: Request):
151     """
152     Protected page - the recall notice records.
153
154     Anyone without a live session is sent back to the login page, which is
155     what makes this a protected route.
156     """
157     user = current_user(request)
158     if not user:
159         return RedirectResponse(url="/login?expired=1", status_code=HTTP_303_SEE_OTHER)
160
161     return templates.TemplateResponse(
162         request,
163         "dashboard.html",
164         {"user": user, "active": "dashboard", "idle_timeout": IDLE_TIMEOUT_SECONDS},
165     )
166
```

Logout (/logout): Destroy the session and redirect the user to the home page.

Screenshot -

As the user has logged out it just redirects to the homepage showing the login button.



Code -

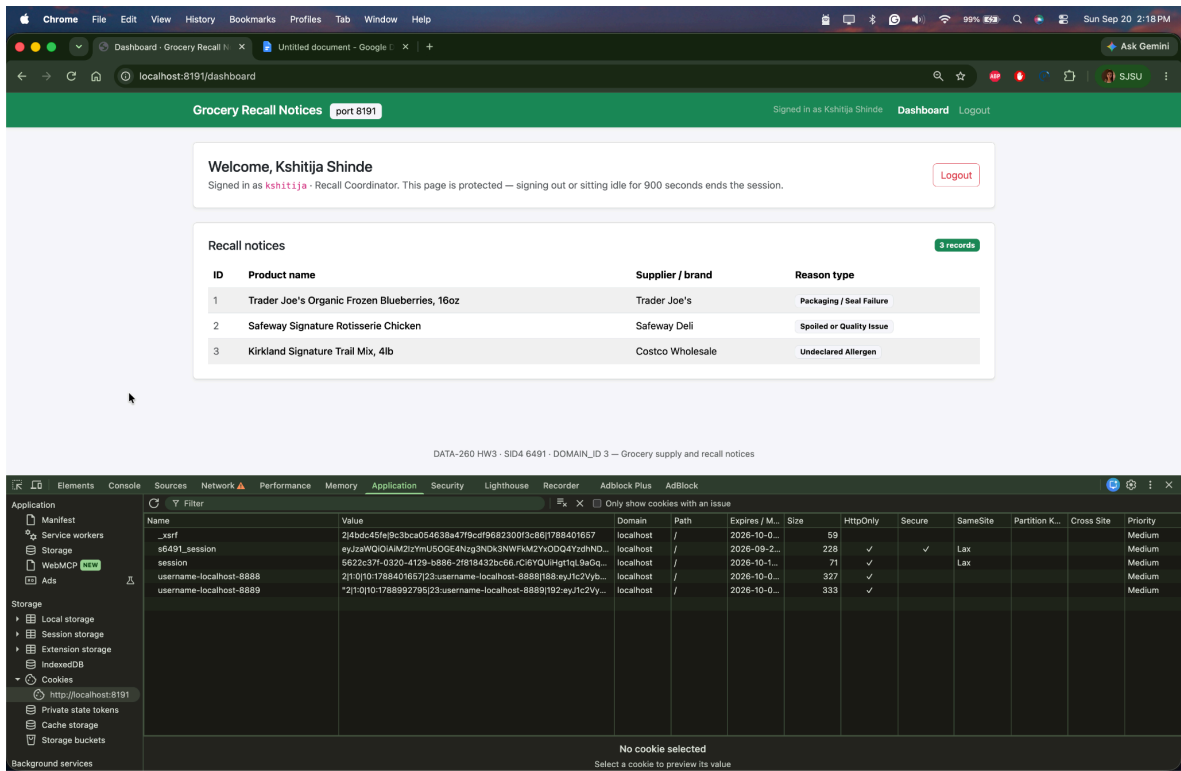
```
base.html U X
code > web_application > templates > base.html > html > body.bg-body-tertiary.d-flex.flex-column.min-vh-100
31     <a class="nav-link {% if active == 'dashboard' %}active fw-semibold{% endif %}"
32         href="/dashboard">Dashboard</a>
33     <a class="nav-link" href="/logout">Logout</a>
34     {% else %}
35     <a class="nav-link {% if active == 'login' %}active fw-semibold{% endif %}"
36         href="/login">Login</a>
37     {% endif %}
38 </div>
39 </div>
40 </div>
41 </nav>
42
43 <main class="container py-4 flex-grow-1">
44     {% block content %}{% endblock %}
45 </main>
46
47 <footer class="container py-4 text-center text-body-secondary small">
48     DATA-250 HW3 &mdot; SID4 6491 &mdot; DOMAIN_ID 3 &mdash; Grocery supply and recall notices
49 </footer>
50
51 <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>
52 {% block scripts %}{% endblock %}
53 </body>
54 </html>
55
```

```
auth.py 5, U X
code > web_application > routers > auth.py > login_page
167
168 @router.get("/logout")
169 def logout(request: Request):
170     """
171     Log out and go back to the home page.
172
173     The sid is deleted from ACTIVE_SESSIONS first. That is the step that makes
174     the old cookie permanently useless, even if somebody saved a copy of it.
175     """
176     sid = request.session.get("sid")
177     if sid:
178         ACTIVE_SESSIONS.pop(sid, None)
179
180     request.session.clear()
181     return RedirectResponse(url="/", status_code=HTTP_303_SEE_OTHER)
182
```


Session Management

Use Starlette's SessionMiddleware to manage user sessions. Store the logged in user's information in the session.

ScreenShot -

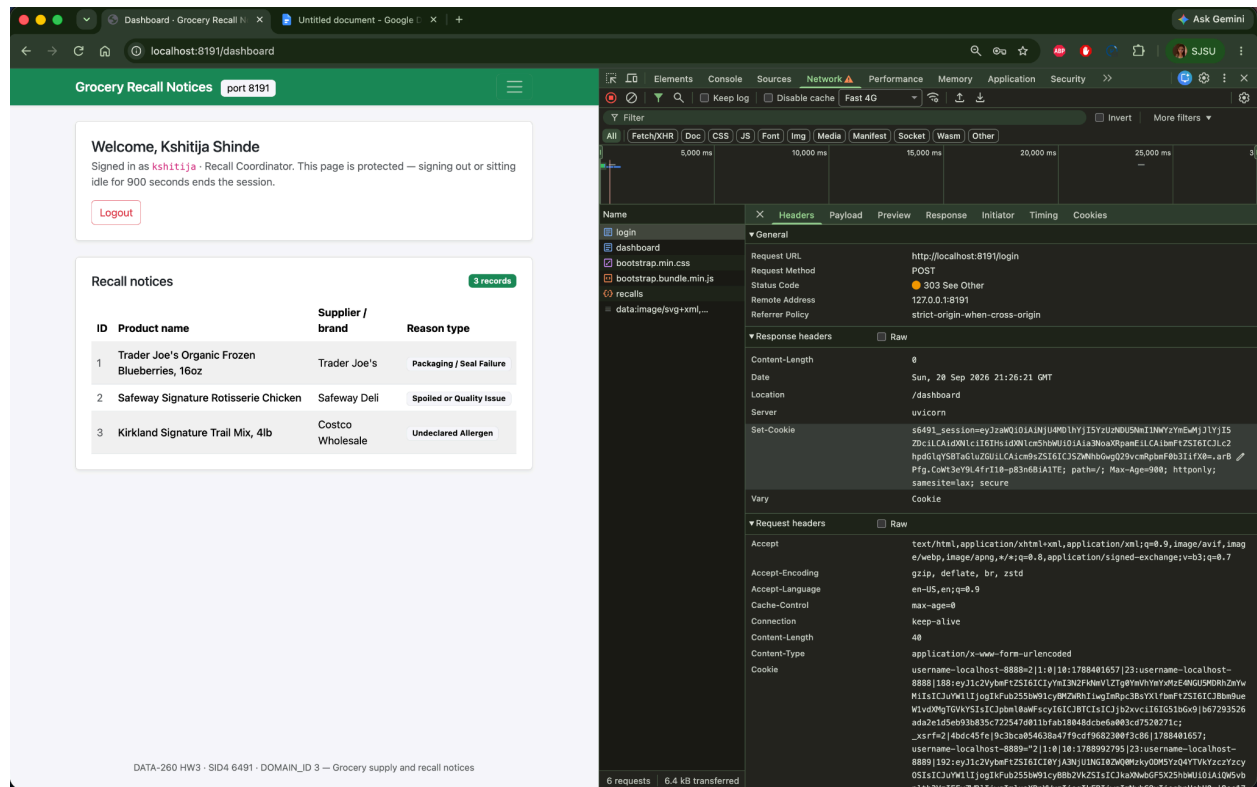


Code -

```
main.py (Working Tree) 6, M X
code > web_application > main.py > ...

1 from fastapi import FastAPI, HTTPException, Response
2 from fastapi.staticfiles import StaticFiles
3 from fastapi.responses import FileResponse
4 from pydantic import BaseModel
5 from typing import List
6
7 import uvicorn
8
9 # this is my backend for grocery recall notices app.
10 # product name is the main field and the supplier is the second field.
11 # student id 819146491
12 # my student id ends in 6491 so my port number is 8000 + (6491 mod 900) = 8191.
13 PORT_BASE = 8191
14
15 app = FastAPI(title="Grocery Supply and Recall Notices API", version="1.0.0")
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Screenshot



Code

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main.py (Working Tree) 6, M x
code > web_application > main.py > ...

1 from fastapi import FastAPI, HTTPException, Response
2 from fastapi.staticfiles import StaticFiles
3 from fastapi.responses import FileResponse
4 from pydantic import BaseModel
5 from typing import List
6 import uvicorn
7
8 # this is my backend for grocery recall notices app.
9 # product name is the main field and the supplier is the second field.
10 # student id 819146491
11 # my student id ends in 6491 so my port number is 8080 + (6491 mod 900) = 8191.
12 PORT_BASE = 8191
13
14 app = FastAPI(title="Grocery Supply and Recall Notices API", version="1.0.0")
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```

Implement an idle session timeout. Prove that an expired or logged out session cannot be reused to reach /dashboard.

Code -

```
auth.py 5, U X
code > web_application > routers > auth.py > ...
52
53 ACTIVE_SESSIONS: dict[str, float] = {}
54
55 IDLE_TIMEOUT_SECONDS = int(os.getenv("SESSION_MAX_AGE", "900"))
56
57
58 def current_user(request: Request):
59     """
60     Return the logged-in user, or None.
61
62     Returns None when any of these is true:
63     - the browser sent no session cookie at all
64     - the cookie's sid is not in ACTIVE_SESSIONS (i.e. it was logged out)
65     - the session has been idle for longer than IDLE_TIMEOUT_SECONDS
66
67     Every successful call refreshes the idle clock, so the timeout measures
68     time since the last request rather than time since login.
69     """
70     sid = request.session.get("sid")
71     if not sid:
72         return None
73
74     last_seen = ACTIVE_SESSIONS.get(sid)
75     if last_seen is None:
76         request.session.clear()
77         return None
78
79     if time.time() - last_seen > IDLE_TIMEOUT_SECONDS:
80         ACTIVE_SESSIONS.pop(sid, None)
81         request.session.clear()
82         return None
83
84     ACTIVE_SESSIONS[sid] = time.time()
85     return request.session.get("user")
86
```

```
auth.py 5, U X
code > web_application > routers > auth.py > ...
167
168 @router.get("/logout")
169 def logout(request: Request):
170     """
171     Log out and go back to the home page.
172
173     The sid is deleted from ACTIVE_SESSIONS first. That is the step that makes
174     the old cookie permanently useless, even if somebody saved a copy of it.
175     """
176     sid = request.session.get("sid")
177     if sid:
178         ACTIVE_SESSIONS.pop(sid, None)
179
180     request.session.clear()
181     return RedirectResponse(url="/", status_code=HTTP_303_SEE_OTHER)
182
```

```
code
File Edit Selection View Go Run Terminal Window Help
data260-6491
Source Control
Changes
Commit
Changes
compare_chunking.py code/rag U
requirements.txt code/rag U
ALUSE.md reports/hw03 U
METRICS.md reports/hw03 U
RUN_LOG.txt reports/hw03 U
verification.json reports/hw03 U
chunk_stats.csv reports/hw03/raw U
retrieval_results.csv reports/hw03/raw U
recompute_tables.py scripts U
verify_hw03.py scripts U
Graph
main* 0 0 0
Problems Output Debug Console Terminal Ports Comments
Terminal
(data260) kshiti@arohandeshmukh@kshiti@MacBook-Air data260-6491 % conda activate data260
cd /Users/kshiti@arohandeshmukh/Downloads/Fall2026/Data260_AgentICAI/Homeworks/Github Repo Location For All Homeworks/data260-6491"
python scripts/prove_session_security.py

=====
POST /login -> HTTP 303
Location -> /dashboard

Set-Cookie: s6491_session=eyJ2aWQ1OjAlZjVhZjA4MjY5NDdJMTg0ODU4OTc2NDI3ODU4MjY1LCA1dWVlc1I6THsldWVlc1c3SHhWUjI0IAA3MoaXpamEjLCA1bmF1ZS16IjI3I2hpdiQ1Y5bTaGluZGJ1LCA1cm9sZS16IjI3I2Mhbgw
g029vcvRpbmF0b3l1TX0=.arCMBa.Ufy09Qvfgfl.c15ei_l9vLC5nH; path=/; Max-Age=5; httponly; samesite=lax; secure

[PASS] HttpOnly attribute present
[PASS] Secure attribute present
[PASS] SameSite attribute present

Keeping a raw copy of the cookie, as an attacker would:
s6491_session=eyJ2aWQ1OjAlZjVhZjA4MjY5NDdJMTg0ODU4OTc2NDI3ODU4MjY1LCA1dWVlc1I6THsldWVlc1c3SHhWUjI0IAA3MoaXpamEjLCA1bmF1ZS16IjI3I2hpdiQ1Y5bTaGluZGJ1LCA1cm9sZS16IjI3I2Mhbgw

=====
CHECK 2 - the session works while it is alive
=====
GET /dashboard with a live session -> HTTP 200
page greets the user by name -> yes
[PASS] logged-in user can reach the dashboard

=====
CHECK 3 - a LOGGED-OUT cookie cannot be replayed
=====
GET /logout -> HTTP 303
GET /dashboard replaying the pre-logout cookie -> HTTP 303
Location -> /loginexpired=1
[PASS] logged-out cookie is rejected

=====
CHECK 4 - an IDLE session expires after 5s
=====
Signed in again -> HTTP 303
GET /dashboard immediately -> HTTP 200 (expected 200)

Sitting idle for 8 seconds, sending no requests at all...
GET /dashboard after idling -> HTTP 303
Location -> /loginexpired=1
[PASS] idle-expired session is rejected

=====
SUMMARY
=====
[PASS] login redirects to /dashboard
[PASS] HttpOnly attribute present
[PASS] Secure attribute present
[PASS] SameSite attribute present
[PASS] logged-in user reaches /dashboard
[PASS] logged-out cookie cannot reach /dashboard
[PASS] fresh session reaches /dashboard
[PASS] idle-expired session cannot reach /dashboard

8/8 checks passed.

test server stopped.
(data260) kshiti@arohandeshmukh@kshiti@MacBook-Air data260-6491 %
```

Styling with Bootstrap: Explore and use Bootstrap to style all pages. Make the application responsive and visually appealing. Use Bootstrap components, such as the Navbar for navigation, Cards for forms and content, Buttons for actions, and Alerts for messages.

-> This requirement has been addressed as per all the above screenshots.

[Auth.py](#) Code ->

```
auth.py 8, U X
code > web_application > routers > auth.py > ...
1  """
2  routers/auth.py - DATA-260 HW3 Part 1
3
4  Every login / logout / dashboard route for my grocery recall app lives here,
5  in a router of its own, because the assignment says the routes must be
6  "Handled Separately in a router" rather than sitting inside main.py.
7
8  SID4 6491 | PORT_BASE 8191 | DOMAIN_ID 3 - grocery supply and recall notices
9  """
10
11  import os
12  import time
13  import uuid
14
15  import bcrypt
16  from fastapi import APIRouter, Form, Request
17  from fastapi.responses import RedirectResponse
18  from fastapi.templating import Jinja2Templates
19  from starlette.status import HTTP_303_SEE_OTHER
20
21  router = APIRouter()
22
23  templates = Jinja2Templates(directory="templates")
24
25
26  def _hash(plain: str) -> bytes:
27      return bcrypt.hashpw(plain.encode(), bcrypt.gensalt())
28
29
30  USERS = {
31      "kshitiya": {
32          "password_hash": _hash("Recall@6491"),
33          "name": "Kshitiya Shinde",
34          "role": "Recall Coordinator",
35      },
36      "inspector": {
37          "password_hash": _hash("Inspect@6491"),
38          "name": "Food Safety Inspector",
39          "role": "Inspector",
40      },
41  }
42
43
44  def verify_password(username: str, password: str) -> bool:
45      """True only when the username exists and the password matches its hash."""
46      account = USERS.get(username)
47      if account is None:
48          bcrypt.checkpw(b"dummy", _hash("dummy"))
49          return False
50      return bcrypt.checkpw(password.encode(), account["password_hash"])
51
52
53  ACTIVE_SESSIONS: dict[str, float] = {}
54
55  IDLE_TIMEOUT_SECONDS = int(os.getenv("SESSION_MAX_AGE", "900"))
56
57
58  def current_user(request: Request):
59      """
60      Return the logged-in user, or None.
61
62      Returns None when any of these is true:
63      - the browser sent no session cookie at all
64      - the cookie's sid is not in ACTIVE_SESSIONS (i.e. it was logged out)
65      - the session has been idle for longer than IDLE_TIMEOUT_SECONDS
66
67      Every successful call refreshes the idle clock, so the timeout measures
68      time since the last request rather than time since login.
69      """
70      sid = request.session.get("sid")
71      if not sid:
72          return None
```

auth.py 5, U X

code > web_application > routers > auth.py > ...

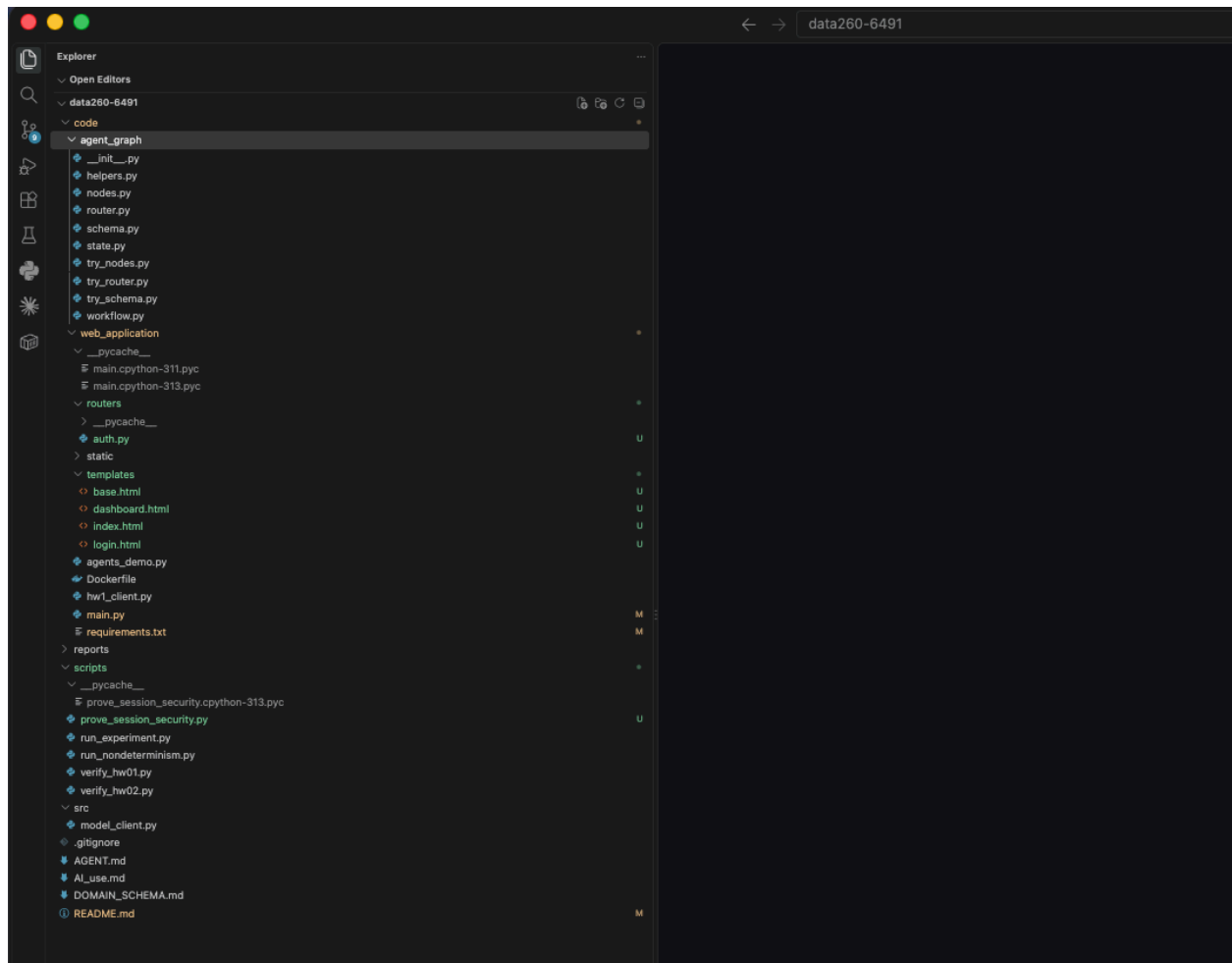
```
66
67     Every successful call refreshes the idle clock, so the timeout measures
68     time since the last request rather than time since login.
69     """
70     sid = request.session.get("sid")
71     if not sid:
72         return None
73
74     last_seen = ACTIVE_SESSIONS.get(sid)
75     if last_seen is None:
76         request.session.clear()
77         return None
78
79     if time.time() - last_seen > IDLE_TIMEOUT_SECONDS:
80         ACTIVE_SESSIONS.pop(sid, None)
81         request.session.clear()
82         return None
83
84     ACTIVE_SESSIONS[sid] = time.time()
85     return request.session.get("user")
86
87
88 @router.get("/")
89 def home(request: Request):
90     """
91     Home page.
92
93     Shows a welcome message for my domain. Offers a Login link when nobody is
94     signed in, and Dashboard + Logout links once somebody is.
95     """
96     return templates.TemplateResponse(
97         request, "index.html", {"user": current_user(request), "active": "home"}
98     )
99
100
101 @router.get("/login")
102 def login_page(request: Request):
103     """Show the empty login form. Already signed in? Go straight to the dashboard."""
104     if current_user(request):
105         return RedirectResponse(url="/dashboard", status_code=HTTP_303_SEE_OTHER)
106
107     return templates.TemplateResponse(
108         request, "login.html", {"user": None, "active": "login", "error": None}
109     )
110
111
112 @router.post("/login")
113 def login(request: Request, username: str = Form(...), password: str = Form(...)):
114     """
115     Handle the submitted login form.
116
117     Correct credentials -> start a session and redirect to /dashboard.
118     Wrong credentials -> redraw the login page with a Bootstrap alert.
119     """
120     if not verify_password(username.strip(), password):
121         return templates.TemplateResponse(
122             request,
123             "login.html",
124             {
125                 "user": None,
126                 "active": "login",
127                 "error": "Incorrect username or password. Please try again.",
128                 "attempted_username": username,
129             },
130             status_code=401,
131         )
132
133     account = USERS[username.strip()]
134
135     sid = uuid.uuid4().hex
136     ACTIVE_SESSIONS[sid] = time.time()
137
```


auth.py 5, U X

code > web_application > routers > auth.py > ...

```
123         "login.html",
124         {
125             "user": None,
126             "active": "login",
127             "error": "Incorrect username or password. Please try again.",
128             "attempted_username": username,
129         },
130         status_code=401,
131     )
132
133     account = USERS[username.strip()]
134
135     sid = uuid.uuid4().hex
136     ACTIVE_SESSIONS[sid] = time.time()
137
138     request.session.clear()
139     request.session["sid"] = sid
140     request.session["user"] = {
141         "username": username.strip(),
142         "name": account["name"],
143         "role": account["role"],
144     }
145
146     return RedirectResponse(url="/dashboard", status_code=HTTP_303_SEE_OTHER)
147
148
149 @router.get("/dashboard")
150 def dashboard(request: Request):
151     """
152     Protected page - the recall notice records.
153
154     Anyone without a live session is sent back to the login page, which is
155     what makes this a protected route.
156     """
157     user = current_user(request)
158     if not user:
159         return RedirectResponse(url="/login?expired=1", status_code=HTTP_303_SEE_OTHER)
160
161     return templates.TemplateResponse(
162         request,
163         "dashboard.html",
164         {"user": user, "active": "dashboard", "idle_timeout": IDLE_TIMEOUT_SECONDS},
165     )
166
167
168 @router.get("/logout")
169 def logout(request: Request):
170     """
171     Log out and go back to the home page.
172
173     The sid is deleted from ACTIVE_SESSIONS first. That is the step that makes
174     the old cookie permanently useless, even if somebody saved a copy of it.
175     """
176     sid = request.session.get("sid")
177     if sid:
178         ACTIVE_SESSIONS.pop(sid, None)
179
180     request.session.clear()
181     return RedirectResponse(url="/", status_code=HTTP_303_SEE_OTHER)
182
```

Screenshot of my templates directory



PART 2

Part 2: Compare Three LlamaIndex Chunking Techniques (Retrieval-Only RAG)

Implement three chunking techniques in LlamaIndex, build in-memory vector indexes, and compare retrieval quality. You'll print the embeddings and retrieval outputs for a shared query, then argue which technique is best and why.

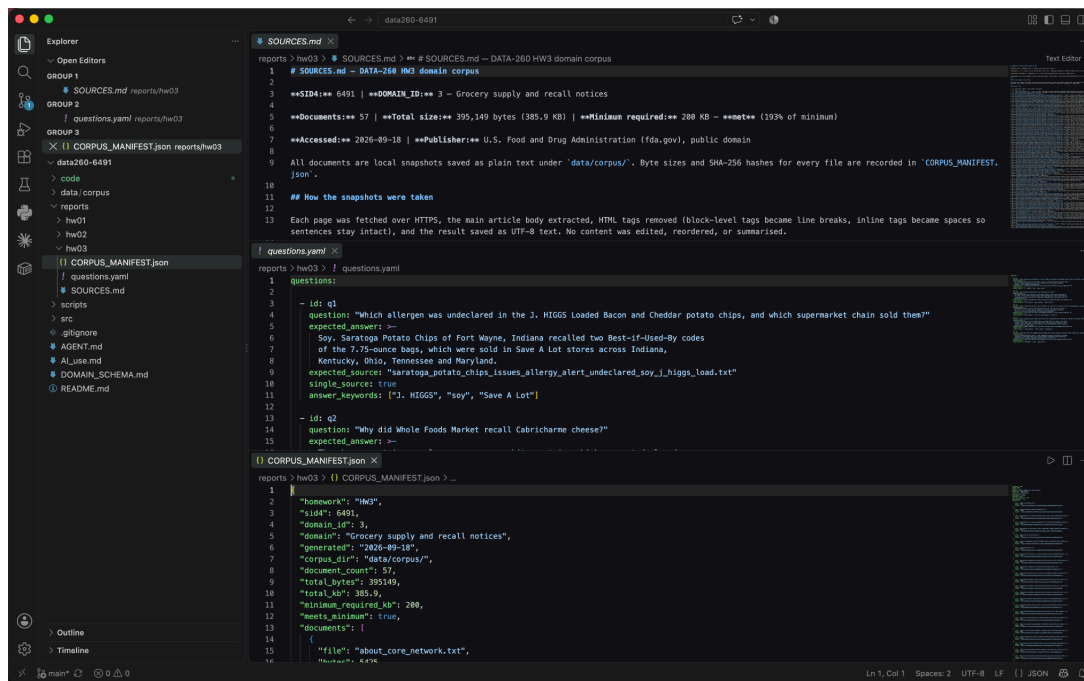
Techniques to implement:

1. Token-based chunking — TokenTextSplitter (LlamaIndex)
2. Semantic chunking — SemanticSplitterNodeParser (LlamaIndex)
3. Sentence-window chunking — SentenceWindowNodeParser (LlamaIndex)

Dataset - Use Tiny Shakespeare (raw text) as a warm-up only:

<https://raw.githubusercontent.com/karpathy/char-rnn/master/data/tinyshakespeare/input.txt>

But your actual dataset (This will be the graded dataset) must be at least 200 KB of local snapshots of public documents relevant to your assigned domain (DOMAIN_ID). Store SOURCES.md with source URLs and access dates, plus CORPUS_MANIFEST.json with each local filename, byte size, and SHA-256 hash.



The screenshot shows a code editor with three files open. The first file, SOURCES.md, contains metadata for a dataset, including a title, domain ID, document count, total size, and access date. The second file, questions.yaml, contains two questions with their expected answers and sources. The third file, CORPUS_MANIFEST.json, is a JSON array containing details for each document in the corpus, such as filename, size, and SHA-256 hash.

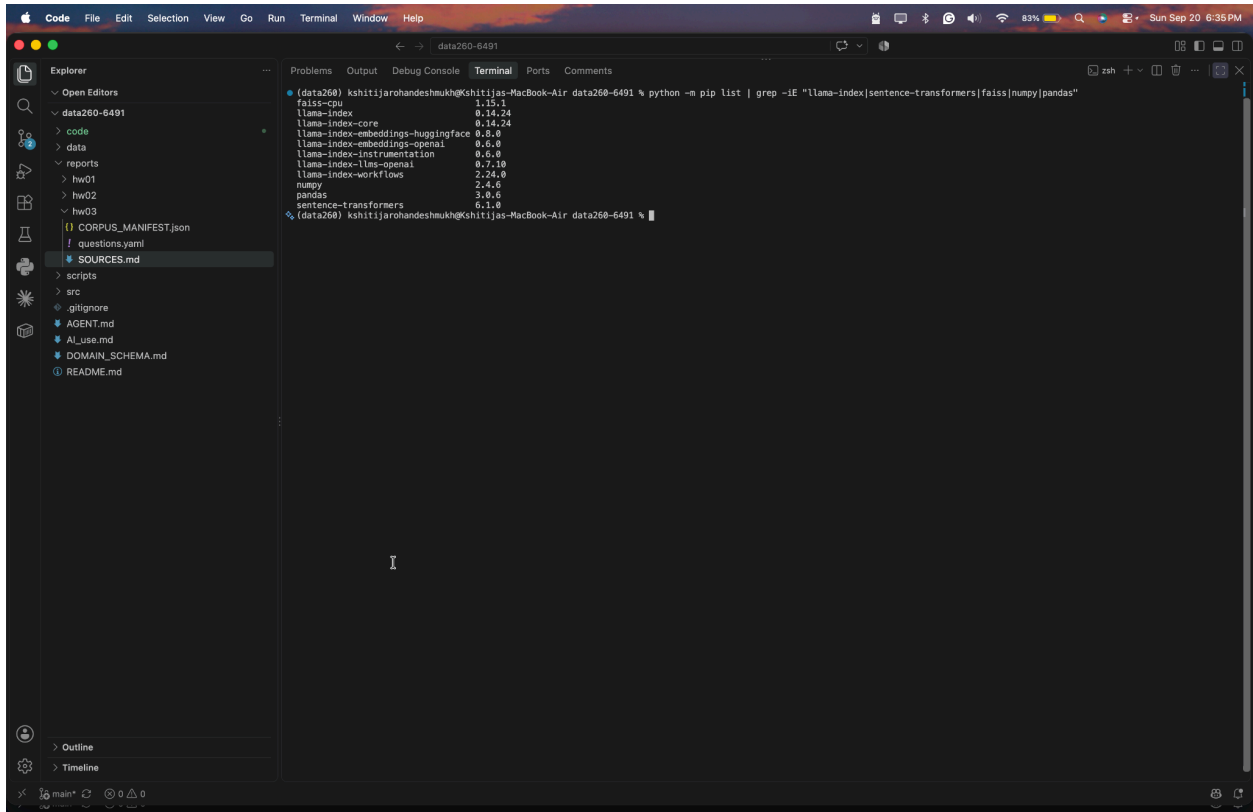
```
reports > hw03 > SOURCES.md > # SOURCES.md - DATA-260 HW3 domain corpus
1 # SOURCES.md - DATA-260 HW3 domain corpus
2
3 ==SID:== 6491 | ==DOMAIN_ID:== 3 - Grocery supply and recall notices
4
5 ==Documents:== 57 | ==Total size:== 395,149 bytes (385.9 KB) | ==Minimum required:== 200 KB - ==met== (139% of minimum)
6
7 ==Accessed:== 2026-09-18 | ==Publisher:== U.S. Food and Drug Administration (fda.gov), public domain
8
9 All documents are local snapshots saved as plain text under 'data/corpus/'. Byte sizes and SHA-256 hashes for every file are recorded in 'CORPUS_MANIFEST.json'.
10
11 ## How the snapshots were taken
12
13 Each page was fetched over HTTPS, the main article body extracted, HTML tags removed (block-level tags became line breaks, inline tags became spaces so sentences stay intact), and the result saved as UTF-8 text. No content was edited, reordered, or summarised.

reports > hw03 > questions.yaml
1 questions:
2
3 - id: q1
4 question: "Which allergen was undeclared in the J. HIGGS Loaded Bacon and Cheddar potato chips, and which supermarket chain sold them?"
5 expected_answer: >=
6 Soy. Saratoga Potato Chips of Fort Wayne, Indiana recalled two Best-If-Used-By codes
7 of the 7.75-ounce bags, which were sold in Save A Lot stores across Indiana,
8 Kentucky, Ohio, Tennessee and Maryland.
9 expected_source: "saratoga_potato_chips_issues_allergy_alert_undeclared_soy_j_higgs_load.txt"
10 single_source: true
11 answer_keywords: ["J. HIGGS", "soy", "Save A Lot"]
12
13 - id: q2
14 question: "Why did Whole Foods Market recall Cabricharme cheese?"
15 expected_answer: >=

reports > hw03 > CORPUS_MANIFEST.json > ...
1 [
2   {
3     "filename": "HW3",
4     "size": 6491,
5     "domain_id": 3,
6     "domain": "Grocery supply and recall notices",
7     "generated": "2026-09-18",
8     "corpus_dir": "data/corpus/",
9     "document_count": 57,
10    "total_bytes": 395149,
11    "total_kb": 385.9,
12    "minimum_required_kb": 200,
13    "meets_minimum": true,
14    "documents": [
15      {
16        "file": "about_core_network.txt",
17        "size": 6491
```

What you must build

- A. Environment & Setup Install: llama-index, llama-index-embeddings-huggingface, sentence-transformers, faiss-cpu, numpy, pandas.
 - a. Use a public sentence embedding model (e.g., sentence-transformers/all-MiniLM-L6-v2). (pull from huggingface)



The screenshot shows a VS Code editor with a terminal window open. The terminal displays the output of the command `python -m pip list | grep -iE "llama-index|sentence-transformers|faiss|numpy|pandas"`. The output lists the installed packages and their versions:

Package	Version
faiss-cpu	1.15.1
llama-index	0.14.24
llama-index-core	0.14.24
llama-index-embeddings-huggingface	0.8.0
llama-index-embeddings-openai	0.6.0
llama-index-instrumentation	0.6.0
llama-index-llms-openai	0.7.10
llama-index-workflows	2.24.0
numpy	2.4.6
pandas	3.0.6
sentence-transformers	6.1.0

The VS Code interface also shows a file explorer on the left with a project structure including files like `CORPUS_MANIFEST.json`, `questions.yaml`, `SOURCES.md`, `scripts`, `src`, `.gitignore`, `AGENT.md`, `AI_use.md`, `DOMAIN_SCHEMA.md`, and `README.md`.

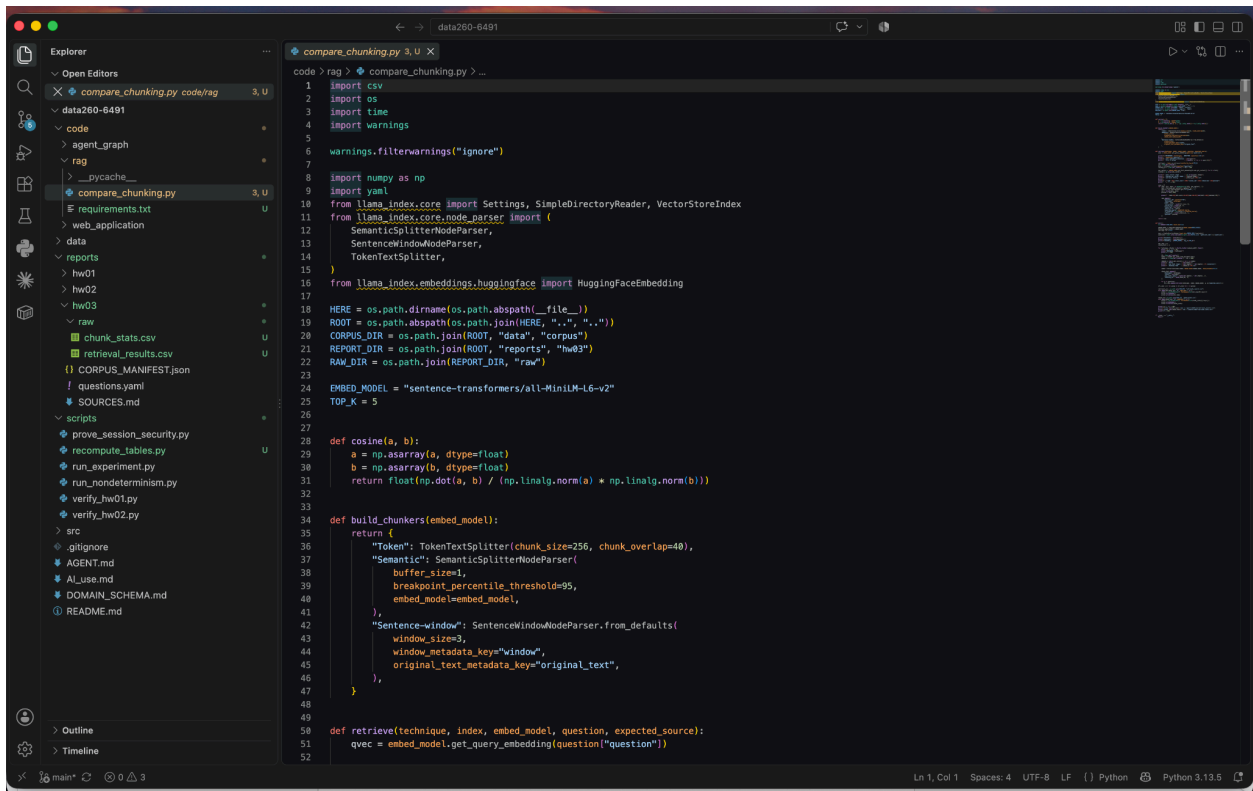
B. One retrieval-only pipeline per technique

For each chunker (Token / Semantic / Sentence-window):

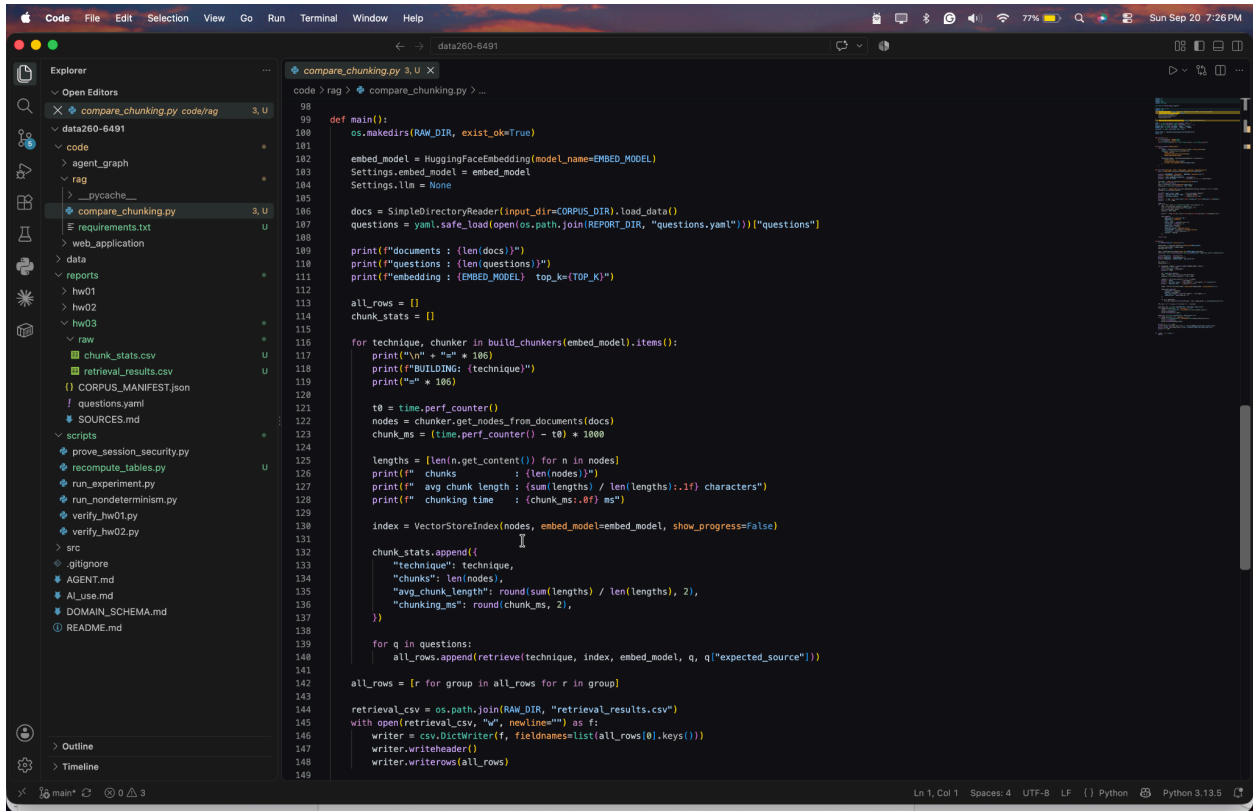
1. Chunking
 - a. Token: set a token `chunk_size` and `chunk_overlap` (choose sensible values). (LlamaIndex)
 - b. Semantic: pick a `buffer_size` and use your embed model for the splitter to find semantically coherent boundaries. (LlamaIndex)
 - c. Sentence-window: split to single sentences and attach a window (neighbor sentences) in metadata to keep surrounding context available. (LlamaIndex)
2. Indexing (in memory)
 - a. Build a `VectorStoreIndex` over your nodes with an in-memory vector store (e.g., `SimpleVectorStore`) to keep everything local and fast. (LlamaIndex)
3. Retrieval-only function - Write a helper that, given a query and `k`, does the following:
 - a. Compute the query embedding (show its dimension and the first 8 values).
 - b. Retrieve top-`k` nodes; for each, compute and print:
 - i. Store similarity score (if available from retriever).
 - ii. Cosine similarity between the query embedding and the document embedding (compute embeddings of the returned chunks explicitly).
 - iii. Chunk length and a short text preview (first ~160 chars).
 - c. Print the shapes of the query vector and the stacked doc vectors.

Your printed output should clearly identify the technique used and list a table with: `rank`, `store_score`, `cosine_sim`, `chunk_len`, `preview`. Save this per-query, per-technique output as machine-readable data (CSV/JSON/JSONL) in `raw/`, along with a script that recomputes the summary tables below from it.

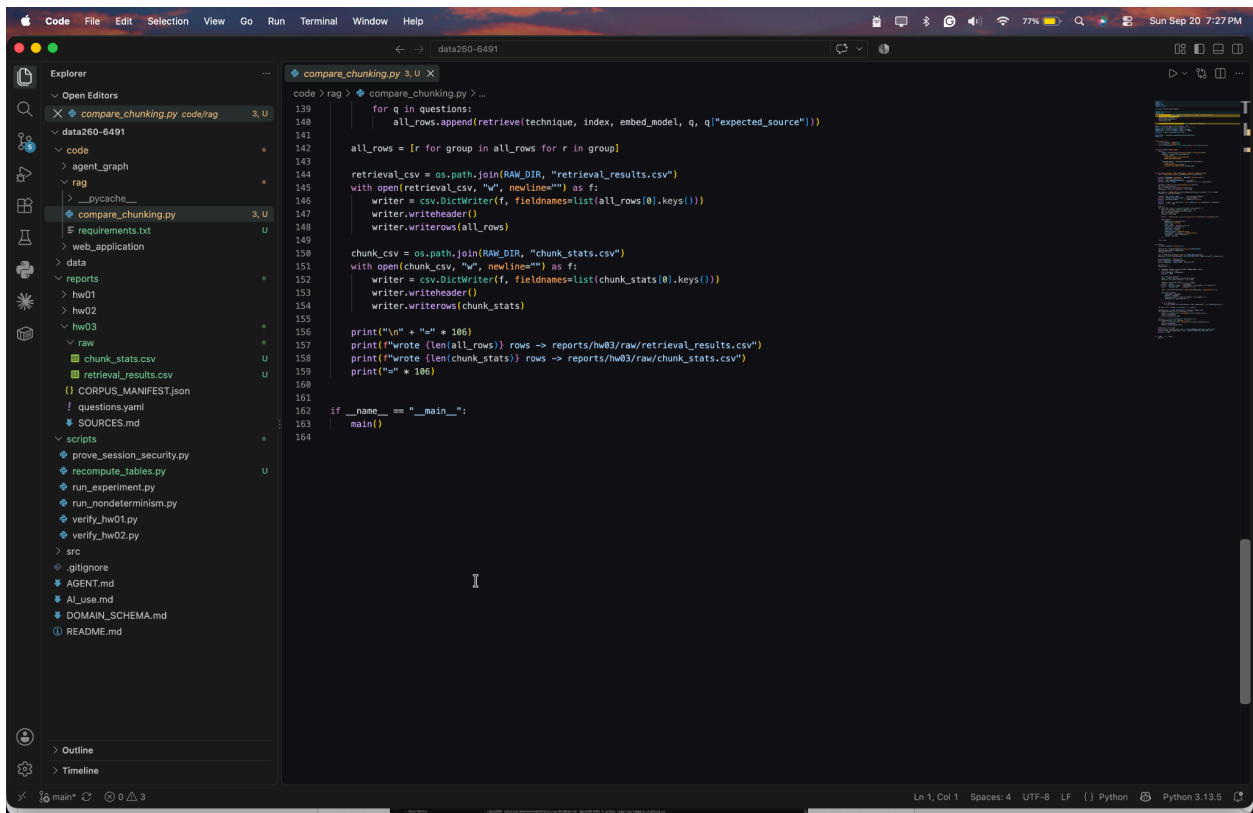
Code -



```
code > rag > compare_chunking.py > ...
1 import csv
2 import os
3 import time
4 import warnings
5
6 warnings.filterwarnings("ignore")
7
8 import numpy as np
9
10 from llama_index.core import Settings, SimpleDirectoryReader, VectorStoreIndex
11 from llama_index.core.node_parser import (
12     SemanticSplitterNodeParser,
13     SentenceWindowNodeParser,
14     TokenTextSplitter,
15 )
16 from llama_index.embeddings.huggingface import HuggingFaceEmbedding
17
18 HERE = os.path.dirname(os.path.abspath(__file__))
19 ROOT = os.path.abspath(os.path.join(HERE, "..", ".."))
20 CORPUS_DIR = os.path.join(ROOT, "data", "corpus")
21 REPORT_DIR = os.path.join(ROOT, "reports", "hw03")
22 RAW_DIR = os.path.join(REPORT_DIR, "raw")
23
24 EMBED_MODEL = "sentence-transformers/all-MiniLM-L6-v2"
25 TOP_K = 5
26
27
28 def cosine(a, b):
29     a = np.asarray(a, dtype=float)
30     b = np.asarray(b, dtype=float)
31     return float(np.dot(a, b) / (np.linalg.norm(a) * np.linalg.norm(b)))
32
33
34 def build_chunkers(embed_model):
35     return {
36         "Token": TokenTextSplitter(chunk_size=256, chunk_overlap=40),
37         "Semantic": SemanticSplitterNodeParser(
38             buffer_size=1,
39             breakpoint_percentile_threshold=95,
40             embed_model=embed_model,
41         ),
42         "Sentence-window": SentenceWindowNodeParser.from_defaults(
43             window_size=3,
44             window_metadata_key="window",
45             original_text_metadata_key="original_text",
46         ),
47     }
48
49
50 def retrieve(technique, index, embed_model, question, expected_source):
51     qvec = embed_model.get_query_embedding(question["question"])
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```



```
code > rag > compare_chunking.py > ...
98
99
100 os.makedirs(RAW_DIR, exist_ok=True)
101
102 embed_model = HuggingFaceEmbedding(model_name=EMBED_MODEL)
103 Settings.embed_model = embed_model
104 Settings.llm = None
105
106 docs = SimpleDirectoryReader(input_dir=CORPUS_DIR).load_data()
107 questions = yaml.safe_load(open(os.path.join(REPORT_DIR, "questions.yaml"))["questions"])
108
109 print(f"documents : {len(docs)}")
110 print(f"questions : {len(questions)}")
111 print(f"embedding : {EMBED_MODEL} top_k={TOP_K}")
112
113 all_rows = []
114 chunk_stats = []
115
116 for technique, chunker in build_chunkers(embed_model).items():
117     print("\n" + "=" * 100)
118     print(f"BUILDING: {technique}")
119     print("=" * 100)
120
121     t0 = time.perf_counter()
122     nodes = chunker.get_nodes_from_documents(docs)
123     chunk_ms = (time.perf_counter() - t0) * 1000
124
125     lengths = [len(n.get_content()) for n in nodes]
126     print(f"chunks : {len(nodes)}")
127     print(f"avg chunk length : {sum(lengths) / len(lengths):.1f} characters")
128     print(f"chunking time : {chunk_ms:.0f} ms")
129
130     index = VectorStoreIndex(nodes, embed_model=embed_model, show_progress=False)
131
132     chunk_stats.append({
133         "technique": technique,
134         "chunks": len(nodes),
135         "avg_chunk_length": round(sum(lengths) / len(lengths), 2),
136         "chunking_ms": round(chunk_ms, 2),
137     })
138
139     for q in questions:
140         all_rows.append(retrieve(technique, index, embed_model, q, q["expected_source"]))
141
142 all_rows = [r for group in all_rows for r in group]
143
144 retrieval_csv = os.path.join(RAW_DIR, "retrieval_results.csv")
145 with open(retrieval_csv, "w", newline="") as f:
146     writer = csv.DictWriter(f, fieldnames=list(all_rows[0].keys()))
147     writer.writeheader()
148     writer.writerows(all_rows)
149
```



```
139     for q in questions:
140         all_rows.append(retrieve(technique, index, embed_model, q, q["expected_source"]))
141
142 all_rows = [r for group in all_rows for r in group]
143
144 retrieval_csv = os.path.join(RAW_DIR, "retrieval_results.csv")
145 with open(retrieval_csv, "w", newline="") as f:
146     writer = csv.DictWriter(f, fieldnames=list(all_rows[0].keys()))
147     writer.writeheader()
148     writer.writerows(all_rows)
149
150 chunk_csv = os.path.join(RAW_DIR, "chunk_stats.csv")
151 with open(chunk_csv, "w", newline="") as f:
152     writer = csv.DictWriter(f, fieldnames=list(chunk_stats[0].keys()))
153     writer.writeheader()
154     writer.writerows(chunk_stats)
155
156 print("\n" + "=" * 100)
157 print(f"Wrote {len(all_rows)} rows -> reports/hw03/raw/retrieval_results.csv")
158 print(f"Wrote {len(chunk_stats)} rows -> reports/hw03/raw/chunk_stats.csv")
159 print("=" * 100)
160
161 if __name__ == "__main__":
162     main()
163
164
```

OUTPUT -

```
data260-6491
python code/rag/compare_chunking.py
Warning: You are sending unauthenticated requests to the HF Hub. Please set a Hugging Face token to enable higher rate limits and faster downloads.
Loading weights: 100%
documents : 57
questions : 5
embedding : sentence-transformers/all-MiniLM-L6-v2_top_k=5

BUILDING: Token
chunks : 684
avg chunk length : 818.3 characters
chunking time : 386 ms

TECHNIQUE: Token QUESTION: q1
Which allergen was undeclared in the J. HIGGS Loaded Bacon and Cheddar potato chips, and which supermarket chain sold them?
query embedding dimension : 384
first 8 values : [-0.8993, 0.0474, -0.0477, 0.0007, 0.0713, -0.0301, -0.049, 0.0438]
query vector shape : (384,)
stacked doc vector shape : (5, 384)
retrieval latency : 11.47 ms

rank store_score cosine_sim chunk_len preview
1 0.6858 0.6942 821 Potato Chips because they may contain undeclared soy allergen. People
2 0.6517 0.6924 747 top right corner of the bag. No illnesses or injuries have been report
3 0.5365 0.6330 795 More Recalls, Market Withdrawals, & Safety Alerts Summary Company Anno
4 0.5346 0.5444 521 Additional Product Issues Product: Imperial Nougat Co., manufactured th
5 0.4984 0.5840 739 to 9/17/2026 To date, no illnesses related to this product have been r

TECHNIQUE: Token QUESTION: q2
Why did Whole Foods Market recall Cabricharme cheese?
query embedding dimension : 384
first 8 values : [-0.0402, -0.0184, 0.0163, 0.0813, 0.0253, 0.0515, -0.0169, 0.0665]
query vector shape : (384,)
stacked doc vector shape : (5, 384)
retrieval latency : 10.56 ms

rank store_score cosine_sim chunk_len preview
1 0.6818 0.6972 847 severe sensitivity to eggs run the risk of a serious or life-threatening
2 0.6557 0.6538 626 Table 10/15/25 Best Kitchen Supply 413 Morse St., We, Washington, D.C.
3 0.5493 0.5920 756 a recall, including retailers and distributors, to ensure that affects
4 0.5249 0.6669 844 More Recalls, Market Withdrawals, & Safety Alerts Summary Company Anno
5 0.5217 0.5091 872 Produce stored at cold temperatures can be contaminated. For example,
best-by dates on and before April 3, 2025. The recalled product was di

TECHNIQUE: Token QUESTION: q3
Which frozen pasta product sold at Walmart was recalled for possible Listeria contamination, and what size was the package?
query embedding dimension : 384
first 8 values : [-0.8954, 0.0085, -0.0091, 0.047, 0.027, 0.005, 0.0156, 0.1258]
query vector shape : (384,)
stacked doc vector shape : (5, 384)
retrieval latency : 9.08 ms

rank store_score cosine_sim chunk_len preview
1 0.5869 0.5455 815 due to an improperly declared allergen (eggs). People who have an alle
2 0.5833 0.4788 872 Custom Smoking of Monroe, Washington, is a processor of a variety of r
3 0.6130 0.5184 802 or # L0896C with an expiration date of 09/19/2027 or 09/20/2027 both s
4 0.5758 0.5310 511 Produce stored at cold temperatures can be contaminated. For example,
5 0.5715 0.5318 847 severe sensitivity to eggs run the risk of a serious or life-threatening

TECHNIQUE: Token QUESTION: q4
Which imported spice brands were found to be contaminated with Salmonella, and who distributed them?
query embedding dimension : 384
first 8 values : [-0.0135, -0.0291, -0.0215, 0.0722, 0.0502, 0.0155, 0.0292, 0.0392]

write 3 rows -> reports/hw3/raw/retrieval_results.csv
write 3 rows -> reports/hw3/raw/chunk_stats.csv
(data260) kshiti@arohandeshmukh@kshiti@MacBook-Air data260-6491 % python code/rag/compare_chunking.py
```

```
data260-6491
python code/rag/compare_chunking.py
Warning: You are sending unauthenticated requests to the HF Hub. Please set a Hugging Face token to enable higher rate limits and faster downloads.
Loading weights: 100%
documents : 57
questions : 5
embedding : sentence-transformers/all-MiniLM-L6-v2_top_k=5

BUILDING: Token
chunks : 684
avg chunk length : 818.3 characters
chunking time : 386 ms

TECHNIQUE: Token QUESTION: q1
Which allergen was undeclared in the J. HIGGS Loaded Bacon and Cheddar potato chips, and which supermarket chain sold them?
query embedding dimension : 384
first 8 values : [-0.8993, 0.0474, -0.0477, 0.0007, 0.0713, -0.0301, -0.049, 0.0438]
query vector shape : (384,)
stacked doc vector shape : (5, 384)
retrieval latency : 11.47 ms

rank store_score cosine_sim chunk_len preview
1 0.6858 0.6942 821 Potato Chips because they may contain undeclared soy allergen. People
2 0.6517 0.6924 747 top right corner of the bag. No illnesses or injuries have been report
3 0.5365 0.6330 795 More Recalls, Market Withdrawals, & Safety Alerts Summary Company Anno
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TECHNIQUE: Token QUESTION: q2
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first 8 values : [-0.0402, -0.0184, 0.0163, 0.0813, 0.0253, 0.0515, -0.0169, 0.0665]
query vector shape : (384,)
stacked doc vector shape : (5, 384)
retrieval latency : 10.56 ms

rank store_score cosine_sim chunk_len preview
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5 0.5217 0.5091 872 Produce stored at cold temperatures can be contaminated. For example,
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TECHNIQUE: Token QUESTION: q3
Which frozen pasta product sold at Walmart was recalled for possible Listeria contamination, and what size was the package?
query embedding dimension : 384
first 8 values : [-0.8954, 0.0085, -0.0091, 0.047, 0.027, 0.005, 0.0156, 0.1258]
query vector shape : (384,)
stacked doc vector shape : (5, 384)
retrieval latency : 9.08 ms

rank store_score cosine_sim chunk_len preview
1 0.5869 0.5455 815 due to an improperly declared allergen (eggs). People who have an alle
2 0.5833 0.4788 872 Custom Smoking of Monroe, Washington, is a processor of a variety of r
3 0.6130 0.5184 802 or # L0896C with an expiration date of 09/19/2027 or 09/20/2027 both s
4 0.5758 0.5310 511 Produce stored at cold temperatures can be contaminated. For example,
5 0.5715 0.5318 847 severe sensitivity to eggs run the risk of a serious or life-threatening

TECHNIQUE: Token QUESTION: q4
Which imported spice brands were found to be contaminated with Salmonella, and who distributed them?
query embedding dimension : 384
first 8 values : [-0.0135, -0.0291, -0.0215, 0.0722, 0.0502, 0.0155, 0.0292, 0.0392]

rank store_score cosine_sim chunk_len preview
1 0.6588 0.6835 173 Additionally, the FDA has subjected the products to "Detention Without
2 0.6483 0.3976 138 Children younger than five, the elderly, and people with weakened imu
3 0.6319 0.4325 139 Symptoms of Salmonella Salmonella is a group of bacteria that can cau
4 0.6169 0.3973 162 More severe cases of salmonellosis may include a high fever, aches, he
5 0.6156 0.3859 281 Purpose The US Food and Drug Administration (FDA) is advising consumer

TECHNIQUE: Sentence-window QUESTION: q5
What was wrong with WanaBana apple cinnamon fruit puree pouches, and which other brands were recalled for the same problem?
query embedding dimension : 384
first 8 values : [-0.0601, -0.0336, -0.0188, 0.0285, -0.0537, -0.0389, -0.1837, 0.0071]
query vector shape : (384,)
stacked doc vector shape : (5, 384)
retrieval latency : 29.92 ms

rank store_score cosine_sim chunk_len preview
1 0.7287 0.6884 116 WanaBana has agreed to voluntarily recall all WanaBana apple cinnamon
2 0.7237 0.7412 208 The FDA has shared the results with the firm whose representatives are
3 0.6591 0.6296 350 In late 2021, after a recall of Apple Cinnamon Fruit Puree pouches int
4 0.6550 0.5183 257 Before the recall, Austrofood did not appropriately conduct a hazard a
5 0.6184 0.6855 172 (Austrofood), the manufacturer of the apple cinnamon fruit puree pouc

write 3 rows -> reports/hw3/raw/retrieval_results.csv
write 3 rows -> reports/hw3/raw/chunk_stats.csv
(data260) kshiti@arohandeshmukh@kshiti@MacBook-Air data260-6491 %
```


The screenshot shows a VS Code window with a terminal at the bottom. The terminal displays the output of a command: `head -3 reports/hw03/raw/retrieval_results.csv`. The output shows a CSV file with 64 rows and 14 columns. The first row is a header with columns: `question_id`, `technique`, `rank`, `score`, `cosine_sim`, `chunk_len`, `source_file`, `expected_source`, `is_expected`, `latency_ms`, `preview`. The second row is a data row with values: `q1`, `Token`, `1`, `0.68765`, `0.69426`, `821`, `saratoga_potato_chips_issues_allergy_alert_undclared_soy_i_higgs_load.txt`, `saratoga_potato_chips_issues_allergy_alert_undclared_soy_i_higgs_load.txt`, `chips_issues_allergy_alert_undclared_soy_i_higgs_load.txt`, `1`, `11.473`. The third row is another data row with values: `q2`, `Token`, `2`, `0.651742`, `0.69246`, `747`, `saratoga_potato_chips_issues_allergy_alert_undclared_soy_i_higgs_load.txt`, `saratoga_potato_chips_issues_allergy_alert_undclared_soy_i_higgs_load.txt`, `chips_issues_allergy_alert_undclared_soy_i_higgs_load.txt`, `1`, `11.473`.

The screenshot shows a VS Code window with a CSV file open in the editor. The CSV file is named `chunk_stats.csv` and has 4 columns: `technique`, `chunks`, `avg_chunk_length`, and `chunking_ms`. The data rows are:

technique	chunks	avg_chunk_length	chunking_ms
1 Token	694	810.27	305.68
2 Semantic	285	1918.6	6444.42
3 Sentence-window	2525	155.77	75.47

The terminal at the bottom shows the command: `head -3 reports/hw03/raw/retrieval_results.csv`. The output is the same as in the first screenshot.

Code File Edit Selection View Go Run Terminal Window Help

data260-6491

chunk_stats.csv U

reports > hw03 > raw > chunk_stats.csv

	technique	chunks	avg_chunk_length	chunking_ms
1	Token	684	810.27	385.68
2	Semantic	285	1918.6	6444.42
3	Sentence-window	2525	155.77	75.47
4				

chunk_stats.csv

retrieval_results.csv

CORPUS_MANIFEST.json

questions.yaml

SOURCES.md

scripts

src

.gitignore

AGENT.md

AI_use.md

DOMAIN_SCHEMA.md

README.md

Problems Output Debug Console Terminal Ports Comments

```
(data260) kshiti@rohandeshmukh@Kshiti@MacBook-Air data260-6491 % ls -l reports/hw03/raw/  
head -3 reports/hw03/raw/retrieval_results.csv  
wc -l reports/hw03/raw/retrieval_results.csv  
76 reports/hw03/raw/retrieval_results.csv  
question_id,technique,rank,store_score,cosine_sim,chunk_len,source_file,expected_source,is_expected,latency_ms,preview  
q1,Token,1,0.685765,0.694209,821,saratoga_potato_chips_issues_allergy_alert_undeclared_soy_j_higgs_load.txt,saratoga_potato_chips_issu  
11:473,Potato Chips because they may contain undeclared soy allergen. People who have an allergy or severe sensitivity to soy run the risk of serious or life-threatening  
q1,Token,2,0.651742,0.692426,747,saratoga_potato_chips_issues_allergy_alert_undeclared_soy_j_higgs_load.txt,saratoga_potato_chips_issu  
11:473,too right corner of the bag. No illnesses or injuries have been reported to date in connection with this problem. The issue was identified through company review  
76 reports/hw03/raw/retrieval_results.csv  
(data260) kshiti@rohandeshmukh@Kshiti@MacBook-Air data260-6491 %
```

Code File Edit Selection View Go Run Terminal Window Help

data260-6491

retrieval_results.csv U

reports > hw03 > raw > retrieval_results.csv

	question_id	technique	rank	store_score	cosine_sim	chunk_len	source_file	
1	q1	Token	1	0.685765	0.694209	821	saratoga_potato_chips_issues_allergy_alert_undeclared_soy_j_higgs_load.txt	saratoga_potato_chips_issu
2	q1	Token	2	0.651742	0.692426	747	saratoga_potato_chips_issues_allergy_alert_undeclared_soy_j_higgs_load.txt	saratoga_potato_chips_issu
3	q1	Token	3	0.536454	0.632993	795	saratoga_potato_chips_issues_allergy_alert_undeclared_soy_j_higgs_load.txt	saratoga_potato_chips_issu
4	q1	Token	4	0.534635	0.544407	567	fda_advises_consumers_and_retailers_not_eat_serve_or_sell_certain_impe.txt	saratoga_potato_chips_issu
5	q1	Token	5	0.498395	0.583986	739	freshpoint_issues_recall_due_improperly_declared_allergen_egg_chicken.txt	saratoga_potato_chips_issu
6	q2	Token	1	0.681765	0.697227	847	whole_foods_market_issues_allergy_alert_undeclared_egg_cabricharme_che.txt	whole_foods_market_issues
7	q2	Token	2	0.555746	0.553752	628	fda_issues_warning_about_imported_cookware_may_leach_lead_august_2025.txt	whole_foods_market_issues
8	q2	Token	3	0.549288	0.561952	256	fda_calls_food_industry_leaders_strengthen_recall_compliance_and_ensur.txt	whole_foods_market_issues
9	q2	Token	4	0.524892	0.666859	844	whole_foods_market_issues_allergy_alert_undeclared_egg_cabricharme_che.txt	whole_foods_market_issues
10	q2	Token	5	0.521683	0.509857	872	so_delicious_dairy_freer_issues_voluntary_recall_salted_caramel_cluste.txt	whole_foods_market_issues
11	q3	Token	1	0.586912	0.545539	815	freshpoint_issues_recall_due_improperly_declared_allergen_egg_chicken.txt	gias_foods_inc_recalls_bet
12	q3	Token	2	0.583319	0.478803	872	public_health_alert_concerning_possible_listeria_contamination_felix_c.txt	gias_foods_inc_recalls_bet
13	q3	Token	3	0.575998	0.518351	862	gias_foods_inc_recalls_bettergoods_authentic_italian_lemon_alfredo_fet.txt	gias_foods_inc_recalls_bet
14	q3	Token	4	0.575836	0.581037	911	food_safety_tips_consumers_retailers_during_outbreak_foodborne_illness.txt	gias_foods_inc_recalls_bet
15	q3	Token	5	0.57151	0.531031	847	whole_foods_market_issues_allergy_alert_undeclared_egg_cabricharme_che.txt	gias_foods_inc_recalls_bet
16	q4	Token	1	0.652246	0.596838	311	public_health_alert_concerning_recalled_everest_and_maggi_brand_spices.txt	public_health_alert_concer
17	q4	Token	2	0.645383	0.594828	968	public_health_alert_concerning_recalled_everest_and_maggi_brand_spices.txt	public_health_alert_concer
18	q4	Token	3	0.616799	0.626487	846	core_publications.txt	public_health_alert_concer

Problems Output Debug Console Terminal Ports Comments

```
(data260) kshiti@rohandeshmukh@Kshiti@MacBook-Air data260-6491 % ls -l reports/hw03/raw/  
head -3 reports/hw03/raw/retrieval_results.csv  
wc -l reports/hw03/raw/retrieval_results.csv  
76 reports/hw03/raw/retrieval_results.csv  
question_id,technique,rank,store_score,cosine_sim,chunk_len,source_file,expected_source,is_expected,latency_ms,preview  
q1,Token,1,0.685765,0.694209,821,saratoga_potato_chips_issues_allergy_alert_undeclared_soy_j_higgs_load.txt,saratoga_potato_chips_issu  
11:473,Potato Chips because they may contain undeclared soy allergen. People who have an allergy or severe sensitivity to soy run the risk of serious or life-threatening  
q1,Token,2,0.651742,0.692426,747,saratoga_potato_chips_issues_allergy_alert_undeclared_soy_j_higgs_load.txt,saratoga_potato_chips_issu  
11:473,too right corner of the bag. No illnesses or injuries have been reported to date in connection with this problem. The issue was identified through company review  
76 reports/hw03/raw/retrieval_results.csv  
(data260) kshiti@rohandeshmukh@Kshiti@MacBook-Air data260-6491 %
```

```
(data260) kshitijarohandeshmukh@Kshitijas-MacBook-Air data260-6491 % scripts/recompute_tables.py
zsh: permission denied: scripts/recompute_tables.py
(data260) kshitijarohandeshmukh@Kshitijas-MacBook-Air data260-6491 % python scripts/recompute_tables.py
Recomputed from reports/hw03/raw/retrieval_results.csv (75 rows)
5 Questions x 3 techniques x top-5

## Retrieval quality
| Technique | Chunks | Avg chunk length | Top-1 cosine | Mean@k cosine | Recall@k | Mean retrieval latency (ms) |
|-----|-----|-----|-----|-----|-----|-----|
| Token | 684 | 810.3 | 0.6456 | 0.5862 | 100% | 18.12 |
| Semantic | 285 | 1918.6 | 0.6322 | 0.5386 | 88% | 7.39 |
| Sentence-window | 2525 | 155.8 | 0.6564 | 0.5598 | 100% | 28.04 |

Top-1 cosine is the mean across questions of the best cosine in that question's top-k.
Recall@k is the fraction of questions whose expected_source appeared anywhere in the top-k.

## Chunking cost
| Technique | Chunks | Chunking time (ms) |
|-----|-----|-----|
| Token | 684 | 386 |
| Semantic | 285 | 6444 |
| Sentence-window | 2525 | 75 |

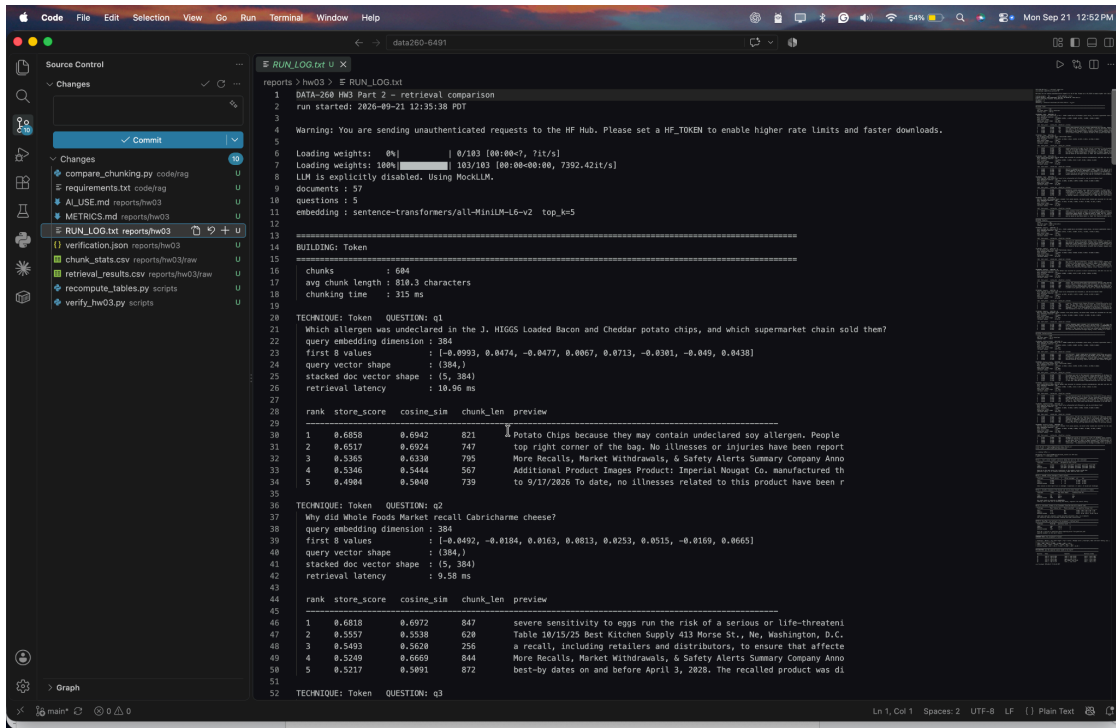
## Per question: was the expected source found in the top-5?
| Question | Token | Semantic | Sentence-window |
|-----|-----|-----|-----|
| q1 | yes (rank 1), top1 0.694 | yes (rank 1), top1 0.676 | yes (rank 1), top1 0.687 |
| q2 | yes (rank 1), top1 0.697 | yes (rank 1), top1 0.687 | yes (rank 1), top1 0.666 |
| q3 | yes (rank 3), top1 0.581 | yes (rank 1), top1 0.587 | yes (rank 1), top1 0.583 |
| q4 | yes (rank 1), top1 0.626 | NO, top1 0.684 | yes (rank 1), top1 0.683 |
| q5 | yes (rank 1), top1 0.629 | yes (rank 1), top1 0.607 | yes (rank 1), top1 0.741 |

(data260) kshitijarohandeshmukh@Kshitijas-MacBook-Air data260-6491 %
```

I've added metrics.md.

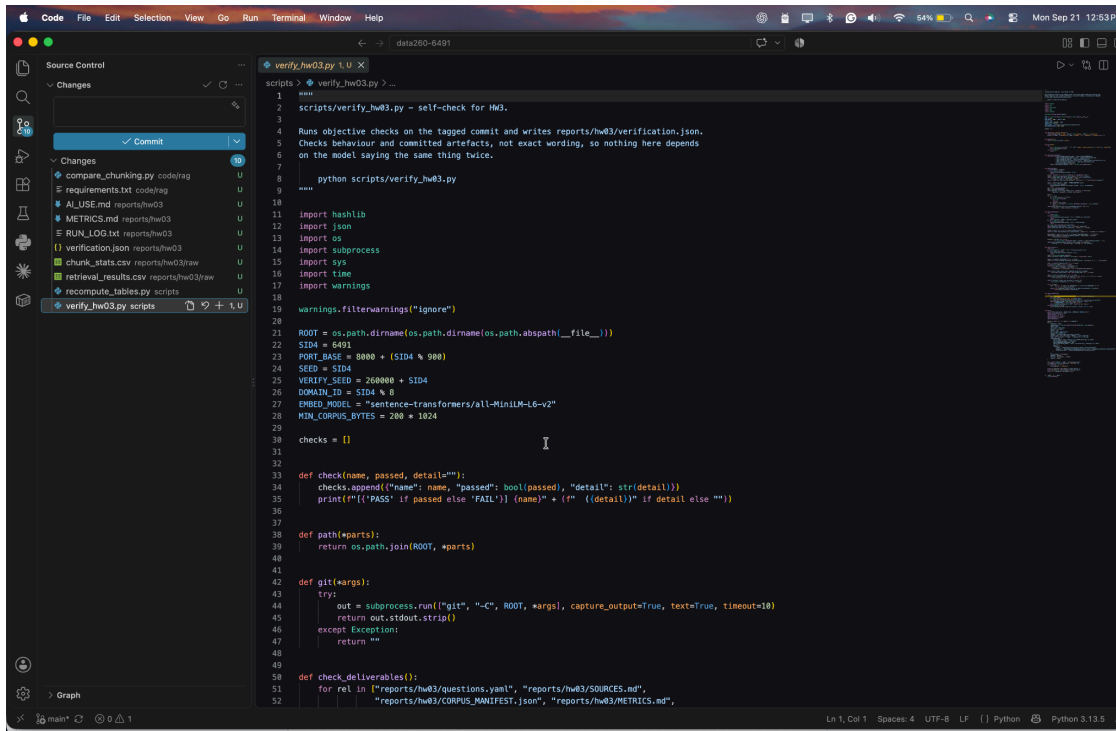
```
reports > hw03 > METRICS.md > ## METRICS.md - DATA-260 HW3 > ## How to reproduce
1 # METRICS.md - DATA-260 HW3
2
3 Every number below was produced by 'scripts/recompute_tables.py' reading
4 'reports/hw03/raw/retrieval_results.csv' and 'chunk_stats.csv'. Nothing was typed by hand.
5
6 ## Configuration
7
8 | | |
9 |---|
10 | SID4 | 6491 |
11 | PORT_BASE | 8191 |
12 | PREFIX | s6491 |
13 | SEED | VERIFY_SEED | 6491 / 266491 |
14 | DOMAIN_ID | 3 (grocery supply and recall notices) |
15 | Hardware | MacBook Air, Apple M4, 16 GB RAM |
16 | OS | macOS (Darwin 25.0.0) |
17 | Python | 3.11.16, conda environment 'data260' |
18 | Embedding model | 'sentence-transformers/all-MiniLM-L6-v2', 384 dimensions, HuggingFace |
19 | Generative model | none - Part 2 is retrieval-only, 'Settings.llm = None' |
20 | Corpus | 57 FDA documents, 395,149 bytes (385.9 KB) |
21 | Questions | 5, of which 4 are single-source |
22 | k | 5 |
23
24 Part 1 is a FastAPI authentication app and uses no model. 'qwen3:8b' via Ollama, the
25 local model from #1 and HW2, is not used anywhere in HW3.
26
27 ## Chunker settings
28
29 | Technique | Class and settings |
30 |-----|-----|
31 | Token | 'TokenTextSplitter(chunk_size=256, chunk_overlap=40)' |
32 | Semantic | 'SemanticSplitterNodeParser(buffer_size=1, breakpoint_percentile_threshold=95)' |
33 | Sentence-window | 'SentenceWindowNodeParser(window_size=3)' |
34
35 ---
36
37 ## 1. Retrieval quality
38
39 | Technique | Chunks | Avg chunk length | Top-1 cosine | Mean@k cosine | Recall@k | Mean retrieval latency (ms) |
40 |-----|-----|-----|-----|-----|-----|-----|
41 | Token | 684 | 810.3 | 0.6456 | 0.5862 | 100% | 9.55 |
42 | Semantic | 285 | 1918.6 | 0.6322 | 0.5386 | 88% | 7.39 |
43 | Sentence window | 2525 | 155.8 | 0.6564 | 0.5598 | 100% | 24.48 |
44
45 Definitions, because two of these are ambiguous in the assignment:
46
47 - **Avg chunk length** is in **characters**.
48 - **Top-1 cosine** is the mean across the five questions of the highest cosine inside
49   that question's top-5, so it reflects consistency rather than one lucky best case.
50 - **Mean@k cosine** averages all 25 cosines per technique (5 questions x 5 ranks).
51 - **Recall@k** is not defined in the assignment. Defined here as the fraction of
52   questions whose 'expected_source' from 'questions.yaml' appeared anywhere in the
```

I've added RUN_LOG.txt



```
reports>hw03> E RUN_LOG.txt
1 DATA-268 HM3 Part 2 - retrieval comparison
2 run started: 2020-09-21 12:35:38 PDT
3
4 Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
5
6 Loading weights: 0% | 0/103 [00:00<7, 7151/s]
7 Loading weights: 100% | 103/103 [00:00<00, 7392.421t/s]
8 LLM is explicitly disabled. Using MockLLM.
9 documents : 57
10 questions : 5
11 embedding : sentence-transformers/all-MiniLM-L6-v2 top_k=5
12
13
14 BUILDING: Token
15
16 chunks : 604
17 avg chunk length : 818.3 characters
18 chunking time : 315 ms
19
20 TECHNIQUE: Token QUESTION: q1
21 Which allergen was undeclared in the J. MIGGS Loaded Bacon and Cheddar potato chips, and which supermarket chain sold them?
22 query embedding dimension : 384
23 first 8 values : [-0.0993, 0.0474, -0.0477, 0.0007, 0.0713, -0.0301, -0.049, 0.0438]
24 query vector shape : (384,)
25 stacked doc vector shape : (5, 384)
26 retrieval latency : 10.96 ms
27
28 rank store_score cosine_sim chunk_len preview
29
30 1 0.6858 0.6942 821 Potato Chips because they may contain undeclared soy allergen. People
31 2 0.6517 0.6924 747 top right corner of the bag. No illnesses or injuries have been report
32 3 0.5365 0.6338 795 More Recalls, Market Withdrawals, & Safety Alerts Summary Company Anno
33 4 0.5346 0.5444 567 Additional Product Images Product: Imperial Nougat Co. Manufactured th
34 5 0.4984 0.5040 739 to 9/17/2026 To date, no illnesses related to this product have been r
35
36 TECHNIQUE: Token QUESTION: q2
37 Why did Whole Foods Market recall Cabricharme cheese?
38 query embedding dimension : 384
39 first 8 values : [-0.0492, -0.0184, 0.0163, 0.0813, 0.0253, 0.0515, -0.0169, 0.0665]
40 query vector shape : (384,)
41 stacked doc vector shape : (5, 384)
42 retrieval latency : 9.58 ms
43
44 rank store_score cosine_sim chunk_len preview
45
46 1 0.6818 0.6972 847 severe sensitivity to eggs run the risk of a serious or life-threateni
47 2 0.5557 0.5538 628 Table 10/15/25 Best Kitchen Supply 413 Morse St., Ne, Washington, D.C.
48 3 0.5493 0.5020 256 a recall, including retailers and distributors, to ensure that affecte
49 4 0.5249 0.6669 844 More Recalls, Market Withdrawals, & Safety Alerts Summary Company Anno
50 5 0.5217 0.5091 872 best-by dates on and before April 3, 2028. The recalled product was di
51
52 TECHNIQUE: Token QUESTION: q3
```

I've added verify_hw03.py



```
scripts>verify_hw03.py>...
1 ==
2 scripts/verify_hw03.py = self-check for HM3.
3
4 Runs objective checks on the tagged commit and writes reports/hw03/verification.json.
5 Checks behaviour and committed artefacts, not exact wording, so nothing here depends
6 on the model saying the same thing twice.
7
8 python scripts/verify_hw03.py
9 ==
10
11 import hashlib
12 import json
13 import os
14 import subprocess
15 import sys
16 import time
17 import warnings
18
19 warnings.filterwarnings("ignore")
20
21 ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
22 SID4 = 5401
23 PORT_BASE = 8000 + (SID4 % 900)
24 SEED = SID4
25 VERIFY_SEED = 260000 + SID4
26 DOMAIN_ID = SID4 % 8
27 EMBED_MODEL = "sentence-transformers/all-MiniLM-L6-v2"
28 MIN_CORPUS_BYTES = 200 * 1024
29
30 checks = []
31
32
33 def check(name, passed, detail=""):
34     checks.append({"name": name, "passed": bool(passed), "detail": str(detail)})
35     print(f"[{'PASS' if passed else 'FAIL'}] (name) + {' (detail)' if detail else ''})")
36
37
38 def path(*parts):
39     return os.path.join(ROOT, *parts)
40
41
42 def git(*args):
43     try:
44         out = subprocess.run(["git", "-C", ROOT, *args], capture_output=True, text=True, timeout=10)
45         return out.stdout.strip()
46     except Exception:
47         return ""
48
49
50 def check_deliverables():
51     for rel in ["reports/hw03/questions.yaml", "reports/hw03/SOURCES.md",
52               "reports/hw03/CORPUS_MAINIFEST.json", "reports/hw03/METRICS.md",
```

Query to use

Before running, create questions.yaml with five realistic domain questions (relevant to your DOMAIN_ID). For each question, record an expected answer and expected source file; at least two questions must depend on information found in only one source document. Commit questions.yaml before adding the retrieval results so the expected answers are not rewritten after seeing the output. What to compare (report section) After you run the three pipelines:

1. Retrieval Quality:
 - a. top-1 cosine (highest similarity among the top-k for that technique)
 - b. mean@k cosine (average of top-k cosines)
 - c. #chunks produced by the chunker and the avg chunk length (characters or tokens)
 - d. retrieval latency in milliseconds (time the similarity search took; simple timer is fine)

Find at least one confidently scored retrieval that does not contain the answer. If your five questions don't produce one, add questions until you find one. Explain why the embedding likely considered it similar.

A confidently scored retrieval that does not contain the answer

Expected source	public_health_alert_concerning_recalled_everest_and_maggi_brand_spices.txt
Returned at rank 1	public_health_advisories_investigations_foodborne_illness_outbreaks.txt
store_score	0.6069
cosine_sim	0.6040
chunk length	887 characters
Expected source anywhere in top-5?	No

For question 4, I asked which imported spice brands had Salmonella and who distributed them. Semantic put this chunk at rank 1 with a score of 0.6069, which is confident, but it's the wrong document and the right one didn't show up in the top 5 at all. The chunk it gave me is just a list of outbreak headlines:

"Listeria - Hispanic-style Cheeses Illnesses - Infused Rice Puffer Fish Poisoning
Illness in Virginia Salmonella - Cashew Cheese from the Cultured Kitchen..."

Embedding likely considered it similar as that chunk is full of the same words as my question. Salmonella is in there several times, plus recall and a bunch of imported foods. The embedding only knows what a chunk is generally about, not which specific thing it names. What my question actually needed was two names, Everest and Amin Trading Agency, and two words aren't enough to beat a chunk that matches on everything else.

```

Code File Edit Selection View Go Run Terminal Window Help
data250-4491
recompute_labels.py
scripts
recompute_labels.py
73: idc = sorted(r["question_id"] for r in rows)
74: table = summarise(rows, chunks)
75: perq = per_question(rows)
76:
77: # "recomputed from reports/two3/rw/retrieval_results.csv" (len(rows) rows)
78: print("Computing {} questions x {}len({}CHUNKS) techniques x {}top-{}".format(
79:     len(rows), len(chunks), len(chunks), len(chunks), len(chunks), len(chunks)))
80: banner1, "top-1 cosine (highest similarity among the topics for that technique)"
81: print("["technique"]>[top-1 cosine] <[per-question best cosine]"
82: print("["technique"]>["top-1 cosine] <[per-question best cosine]"
83: for t in TECHNIQUES:
84:     grouped = by_question(rows, t)
85:     bests = max(chunks["cosine_max"] for x in grouped(x) for x in qids)
86:     detail = " ".join("{}q[{}b:4f]" for q, b in zip(ids, bests))
87:     print("["t"]>[Statistics.mean(bests)]<[6.4f, detail]")
88:     banner1, "Reported as the mean across the 5 questions of the highest cosine inside that"
89:     question's top-1, so it reflects consistency rather than one best case."
90:     print("["technique"]>[Statistics.mean(bests)]<[6.4f, detail]"
91:     banner2, "Month cosine (average of topic cosines)"
92:     print("["technique"]>[Statistics.mean(bests)]<[6.4f, detail]"
93:     for t in TECHNIQUES:
94:         vals = [float(r["cosine_max"]) for r in rows if r["technique"] == t]
95:         print("["t"]>[Statistics.mean(vals)]<[6.4f, detail]"
96:         banner1, "Every cosine in every top-{} chunk is averaged: {}len(chunks) questions x {} ranks x {}len(ids) values per technique."
97:         banner1, "Chunks produced by the lister and the avg chunk lengths (characters or tokens)"
98:         print("["technique"]>[len(chunks)]<[1f] Avg chunk length: {}[len(chunks)]<[1f] Chunks per question: {}[len(chunks)]<[1f]"
99:         for r in rows:
100:             print("["r["technique"]>[len(chunks)]<[1f] Avg chunk length: {}[len(chunks)]<[1f] Chunks per question: {}[len(chunks)]<[1f]"
101:             print("["r["technique"]>[len(chunks)]<[1f] Avg chunk length: {}[len(chunks)]<[1f] Chunks per question: {}[len(chunks)]<[1f]"
102:             print("["r["technique"]>[len(chunks)]<[1f] Avg chunk length: {}[len(chunks)]<[1f] Chunks per question: {}[len(chunks)]<[1f]"
103:             print("["r["technique"]>[len(chunks)]<[1f] Avg chunk length: {}[len(chunks)]<[1f] Chunks per question: {}[len(chunks)]<[1f]"
104:             print("["r["technique"]>[len(chunks)]<[1f] Avg chunk length: {}[len(chunks)]<[1f] Chunks per question: {}[len(chunks)]<[1f]"
105:             print("["r["technique"]>[len(chunks)]<[1f] Avg chunk length: {}[len(chunks)]<[1f] Chunks per question: {}[len(chunks)]<[1f]"
106:             print("["r["technique"]>[len(chunks)]<[1f] Avg chunk length: {}[len(chunks)]<[1f] Chunks per question: {}[len(chunks)]<[1f]"
107:             print("["r["technique"]>[len(chunks)]<[1f] Avg chunk length: {}[len(chunks)]<[1f] Chunks per question: {}[len(chunks)]<[1f]"
108:             print("["r["technique"]>[len(chunks)]<[1f] Avg chunk length: {}[len(chunks)]<[1f] Chunks per question: {}[len(chunks)]<[1f]"
109:             print("["r["technique"]>[len(chunks)]<[1f] Avg chunk length: {}[len(chunks)]<[1f] Chunks per question: {}[len(chunks)]<[1f]"
110:             print("["r["technique"]>[len(chunks)]<[1f] Avg chunk length: {}[len(chunks)]<[1f] Chunks per question: {}[len(chunks)]<[1f]"
111:             print("["r["technique"]>[len(chunks)]<[1f] Avg chunk length: {}[len(chunks)]<[1f] Chunks per question: {}[len(chunks)]<[1f]"
112:             print("["r["technique"]>[len(chunks)]<[1f] Avg chunk length: {}[len(chunks)]<[1f] Chunks per question: {}[len(chunks)]<[1f]"
113:             print("["r["technique"]>[len(chunks)]<[1f] Avg chunk length: {}[len(chunks)]<[1f] Chunks per question: {}[len(chunks)]<[1f]"
114:             print("["r["technique"]>[len(chunks)]<[1f] Avg chunk length: {}[len(chunks)]<[1f] Chunks per question: {}[len(chunks)]<[1f]"
115:             print("["r["technique"]>[len(chunks)]<[1f] Avg chunk length: {}[len(chunks)]<[1f] Chunks per question: {}[len(chunks)]<[1f]"
116:             print("["r["technique"]>[len(chunks)]<[1f] Avg chunk length: {}[len(chunks)]<[1f] Chunks per question: {}[len(chunks)]<[1f]"
117:             print("["r["technique"]>[len(chunks)]<[1f] Avg chunk length: {}[len(chunks)]<[1f] Chunks per question: {}[len(chunks)]<[1f]"
118:             print("["r["technique"]>[len(chunks)]<[1f] Avg chunk length: {}[len(chunks)]<[1f] Chunks per question: {}[len(chunks)]<[1f]"
119:             print("["r["technique"]>[len(chunks)]<[1f] Avg chunk length: {}[len(chunks)]<[1f] Chunks per question: {}[len(chunks)]<[1f]"
120:             print("["r["technique"]>[len(chunks)]<[1f] Avg chunk length: {}[len(chunks)]<[1f] Chunks per question: {}[len(chunks)]<[1f]"
121:             print("["r["technique"]>[len(chunks)]<[1f] Avg chunk length: {}[len(chunks)]<[1f] Chunks per question: {}[len(chunks)]<[1f]"
122:             print("["r["technique"]>[len(chunks)]<[1f] Avg chunk length: {}[len(chunks)]<[1f] Chunks per question: {}[len(chunks)]<[1f]"
123:             print("["r["technique"]>[len(chunks)]<[1f] Avg chunk length: {}[len(chunks)]<[1f] Chunks per question: {}[len(chunks)]<[1f]"
124:             print("["r["technique"]>[len(chunks)]<[1f] Avg chunk length: {}[len(chunks)]<[1f] Chunks per question: {}[len(chunks)]<[1f]"

```

Code
File
Edit
Selection
View
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Run
Terminal
Window
Help
data250-6491

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Open Files
data250-6491
code
agent_graph
img
python
compare_ranking.py
requirements.txt
web_application
data
reports
hw01
hw02
hw03
raw
chunk_data.csv
verified_results.csv
CORPUS_MANIFEST.json
questions.yaml
SOURCES.md
scripts
prove_session_security.py
recompute_tables.py
run_experiment.py
run_xondemosem.py
verify_hw01.py
verify_hw02.py
gitignore
AGENT.md
AGENT_SCHEMA.md
README.md
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Problems
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```

❖ (data260) ksh11@jarandhans-mahagokhlije-MacBook-Air: data260-6491 % clear
❖ (data260) ksh11@jarandhans-mahagokhlije-MacBook-Air: data260-6491 % python scripts/recompute_tables.py
Reported from reports/hw01/run/retrieval_results.csv (75 rows)
5 questions & 3 techniques & top-5

METRIC 1: Top-1 cosine (highest similarity among the top-5 for that technique)
Technique      Top-1 cosine      per-question base cosine
-----
Token          0.4495             0.18-0.092  0.2-0.097  0.18-0.310  0.18-0.254  0.18-0.208
Semantic       0.8322             0.48-0.788  0.24-0.869  0.18-0.571  0.18-0.884  0.18-0.587
Sentence-window 0.1564             0.18-0.075  0.2-0.054  0.18-0.314  0.18-0.871  0.18-0.712

Reported as the mean across the 5 questions of the highest cosine inside that
question's top-5, so it reflects consistency rather than one best case.

METRIC 2: meanq cosine (average of top-4 cosines)
Technique      Meanq cosine      k      values averaged      min      max
-----
Token          0.5862             5      25      0.4788      0.6972
Semantic       0.1386             5      25      0.1201      0.6859
Sentence-window 0.5580             5      25      0.3315      0.7412

Every cosine in every top-5 list is averaged: 5 questions x 5 ranks = 25 values per technique.

METRIC 3: #chunks produced by the chunker and the avg chunk length (characters or tokens)
Technique      Chunks      Avg chunk length      Chunking time (ms)
-----
Token          684      816.3      344
Semantic       280      333.6      644
Sentence-window 2325      335.8      75

Avg chunk length is measured in CHARACTERS.
Chunking time is the cost of building the chunks, separate from search latency.

METRIC 4: retrieval latency in milliseconds (time the similarity search took)
Technique      Mean latency (ms)      Chunks searched      per-question latency (ms)
-----
Token          18-122             684      11.47  18.56  9.60  9.86  9.91
Semantic       7-193             280      18.16  7.76  14.7  6.54  1.58
Sentence-window 28-184             2325      26.95  26.82  26.98  30.33  26.92

Tied with time_perf_counter() around retriever.retrieve() only, so it measures
the similarity search and excludes chunking and index construction.

METRIC 5: Recall@k (not defined in the assignment - defined here)
Technique      Recall@k      questions found      k
-----
Token          184q             5 of 5      5
Semantic       384q             4 of 5      5
Sentence-window 384q             4 of 5      5

Recall@k = fraction of questions whose expected source from questions.yaml
appeared anywhere in the top-5 results.

COMBINED TABLE (the assignment's format)

```

CodeFileEditSelectionViewGoRunTerminalWindowHelp

data260-6491

zsh

Explorer

data260-6491

code

agent_graph

rag

__pycache__

compare_chunking.py

requirements.txt

web_application

data

reports

hw01

hw02

hw03

raw

chunk_stats.csv

retrieval_results.csv

CORPUS_MANIFEST.json

questions.yaml

SOURCES.md

scripts

prove_session_security.py

recompute_tables.py

run_experiment.py

run_nondeterminism.py

verify_hw01.py

verify_hw02.py

src

.gitignore

AGENT.md

AI_use.md

DOMAIN_SCHEMA.md

README.md

Outline

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Output

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Comments

(data260) kshiti@rohanandeshmukh@kshiti@MacBook-Air data260-6491 % python scripts/recompute_tables.py

Technique Meank cosine k values averaged min max

Token 0.5862 5 25 0.4788 0.6972

Semantic 0.5386 5 25 0.3781 0.6869

Sentence-window 0.5588 5 25 0.3315 0.7412

Every cosine in every top-5 list is averaged: 5 questions x 5 ranks = 25 values per technique.

METRIC 3: #chunks produced by the chunker and the avg chunk length (characters or tokens)

Technique Chunks Avg chunk length Chunking time (ms)

Token 684 810.3 386

Semantic 285 1918.6 6444

Sentence-window 2525 155.8 75

Avg chunk length is measured in CHARACTERS.

Chunking time is the cost of building the chunks, separate from search latency.

METRIC 4: retrieval latency in milliseconds (time the similarity search took)

Technique Mean latency (ms) Chunks searched per-question latency (ms)

Token 18.12 684 11.47 18.56 9.68 9.06 9.91

Semantic 7.93 285 10.14 7.96 7.42 6.64 7.59

Sentence-window 28.04 2525 26.95 28.02 24.98 30.33 29.92

Timed with time.perf_counter() around retriever.retrieve() only, so it measures the similarity search and excludes chunking and index construction.

METRIC R: Recall@k (not defined in the assignment - defined here)

Technique Recall@k questions found k

Token 100% 5 of 5 5

Semantic 88% 4 of 5 5

Sentence-window 100% 5 of 5 5

Recall@k = fraction of questions whose expected_source from questions.yaml appeared anywhere in the top-5 results.

COMBINED TABLE (the assignment's format)

Technique | Chunks | Avg chunk length | Top-1 cosine | Meank cosine | Recall@k | Mean retrieval latency (ms) |

Token | 684 | 810.3 | 0.6456 | 0.5862 | 100% | 18.12 |

Semantic | 285 | 1918.6 | 0.6322 | 0.5386 | 88% | 7.93 |

Sentence-window | 2525 | 155.8 | 0.6564 | 0.5588 | 100% | 28.04 |

PER QUESTION: was the expected source found in the top-5?

Question Token Semantic Sentence-window

q1 rank 1, top1 0.694 rank 1, top1 0.676 rank 1, top1 0.687

q2 rank 1, top1 0.697 rank 1, top1 0.687 rank 1, top1 0.666

q3 rank 3, top1 0.591 rank 1, top1 0.587 rank 1, top1 0.583

q4 rank 1, top1 0.626 NOT FOUND, top1 0.604 rank 1, top1 0.683

q5 rank 1, top1 0.629 rank 1, top1 0.607 rank 1, top1 0.741

2. Observations (1–2 short paragraphs):

- Discuss why one technique performed better on this query (e.g., sentence coherence, semantic boundary detection, token-budget alignment, context carried via sentence window).
- If the best technique differs across your optional extra queries, mention it.

For me, the context carried via the sentence window worked best. It put the right document at rank 1 for all five questions, while Token got four out of five and Semantic got four and missed one completely. I think this is because its chunks are tiny. When a recall notice says the thing you're looking for, that whole chunk is just that fact and nothing else.

You can see it on q5 where the sentence window scored 0.7412 and the other two only got 0.6290 and 0.6073. My corpus data is FDA recall notices, which are written in short plain sentences, and most of my questions are asking for one specific name like a brand or a distributor. So small chunks suit it.

Semantic did the worst and I think its own settings caused that. It only made 205 chunks and they were massive in length. On q4 it never found the actual spice alert at all. That chunk had loads of words like Salmonella and recall in it, so it looked similar, but it didn't contain the answer. It also had the lowest Mean@k cosine at 0.5386, so it wasn't just one bad question.

One thing I noticed is that the best technique wasn't the same every time. If you only look at the top 1 cosine, Token actually won three of the five questions, Semantic won q3 and sentence window won q5. Token also had the best Mean@k at 0.5862, so its medium sized chunks scored more evenly down the whole list, while sentence-window was higher at rank 1 but dropped off faster after that.

3. Your conclusion (2–5 sentences): State which technique you judge best for this corpus and why supported by your measurements.

I think the sentence window is the best one for this corpus. It was the only technique that found the right document at rank 1 every single time, and it had the highest top-1 cosine at 0.6564. The downside is speed, since 2,525 chunks made searching take 24.48 ms compared to Token's 9.55 ms, although it was actually the fastest to chunk at only 73 ms versus Semantic's 6,357 ms.

Token is a really close second and honestly a safer pick if the corpus got much bigger, because it also got 100% Recall@5, had the best Mean@k at 0.5862, and was about two and a half times faster to search. I wouldn't use Semantic here, because it was the only one that missed a question, scored lowest on both cosine measures, and took twenty times longer to chunk than Token.